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UG Program in Artificial Intelligence and Data Science

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SHAH & ANCHOR KUTCHHI ENGINEERING COLLEGE

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Faculty of Engineering & Technology | Department of Artificial Intelligence and Data Science

TECHNICAL MICRO-PROJECT REPORT

on

FAIR CREDIT: ALGORITHMIC BIAS AUDITING AND MITIGATION IN DIGITAL MICRO-LENDING

Course: **Trustworthy & Responsible AI (ADCOR1PC303)**

Submitted by

Ritish Oswal

Roll no. 34

Under the guidance of

Milind Khairnar sir

Academic Year 2026–2027

Semester V

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TECHNICAL MICRO-PROJECT REPORT

Fair Credit: Algorithmic Bias Auditing and Mitigation in Digital Micro-Lending

Submitted in partial fulfilment of the requirements of the Continuous Internal Assessment for Practical (CIAP, 25 Marks) for the course ***Trustworthy & Responsible AI (ADCOR1PC303)****, Department of Artificial Intelligence and Data Science*

Submitted by

Ritish Oswal

Roll no. 34

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CERTIFICATE

This is to certify that the Technical Micro-Project Report entitled "**Fair Credit: Algorithmic Bias Auditing and Mitigation in Digital Micro-Lending**" is a bonafide record of work carried out by Ritish Oswal, Roll No. 34, of the Department of Artificial Intelligence and Data Science, Shah & Anchor Kutchhi Engineering College, in partial fulfilment of the Continuous Internal Assessment for Practical (CIAP) for the course *Trustworthy & Responsible AI (ADCOR1PC303)* during the academic year 2026–2027.

The report is submitted for evaluation against the CIAP rubric. Approval and signatures are to be completed by the authorised evaluators.

Guide signature: _____
Head of Department signature: _____
Guide Head of Department
Guide name/designation: Milind Khairnar sir HOD name: Dr. Pravin Shinde sir

Date: 30 / 10 / 2026

Place: Mumbai

STUDENT DECLARATION & ANTI-PLAGIARISM UNDERTAKING

We declare that this written submission represents our own ideas and work, expressed in our own words. Where others' ideas or words have been included, we have adequately cited and referenced the original sources in IEEE format. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated or falsified any idea, data, fact or source in this submission.

We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources that have not been properly cited or from whom proper permission has not been taken.

We further declare that the dataset used in this project is synthetic and generated by the source code in Appendix A. No personal data of any real individual was collected or processed.

Ritish Oswal

Roll No. - 34

Signature _____

Date: 30/ 10 / 2026

COURSE ARTICULATION & OUTCOME ALIGNMENT SHEET

NBA SAR Criterion 3: mapping of the micro-project to Course Outcomes (CO), Program Outcomes (PO) and Program Specific Outcomes (PSO) for *Trustworthy & Responsible AI (ADCOR1PC303)*.

Project activity component	Mapped CO	Bloom's level	Mapped POs / PSOs	Measurable benchmark	Evidence in this project
Feature attribution and proxy analysis (permutation importance, LR coefficients, gender-leakage AUC)	**CO 1**	BL 2 / BL 3	PO1, PO2, PO5, PSO1	Attributions computed for every feature with no error	Table 5.4; leakage AUC 0.897
Fairness audit and debiasing pipeline (reweighing, equal-opportunity and DIR thresholds)	**CO 2**	BL 3 / BL 4	PO1, PO2, PO3, PO5, PSO1	DIR ≥ 0.80; TPR gap minimised towards 0.00	DIR 0.718 → 0.909–0.980; |EOD| 0.164 → 0.011–0.051
Robustness stress test (input noise) and threat assessment	**CO 3**	BL 4 / BL 5	PO1, PO2, PO3, PO5, PSO2	Fairness gain maintained under perturbation	Table 6.1 (noise only; FGSM, MIA and DP not in scope)
Legal compliance audit and model card (DPDP, RBI, EU AI Act, ISO)	**CO 3**	BL 4 / BL 5	PO2, PO6, PO7, PO8, PSO3	Model card aligned with EU AI Act Annex III	Table 6.3; Appendix B
End-to-end reproducible pipeline and report	**CO 1 – CO 3**	BL 5 / BL 6	PO2, PO3, PO5, PO9, PSO3	One-command run; live dashboard (future scope)	Appendix A; dashboard not deployed

CIAP 25-MARK RUBRIC EVALUATION SHEET

Continuous Internal Assessment for Practical (CIAP) • Trustworthy & Responsible AI (ADCOR1PC303)

Criteria (weight)	Excellent (4)	Good (3)	Satisfactory (2)	Needs improvement (1)
Problem Definition & Scope (15%)	Clearly defined problem with a focused and achievable scope.	Problem identified, but scope may be unclear or too broad/narrow.	Problem vaguely defined, or scope missing.	Problem not clearly identified or missing.
Design & Methodology (20%)	Innovative and well-structured approach with justification for chosen methods.	Appropriate design and methodology, but limited justification.	Basic design and methodology, with some justification.	Weak or missing design and methodology.
Implementation & Results (25%)	Project completed effectively with clear and well-presented results.	Project completed, but results unclear or poorly presented.	Project partially completed with limited or unclear results.	Project not completed or results missing.
Analysis & Interpretation (20%)	Thorough analysis with insightful interpretation of results.	Analysis attempts were made, but interpretation could be improved.	Basic analysis with limited interpretation.	Weak or missing analysis and interpretation.
Report & Presentation (20%)	Clear, concise, and well-organized report with effective presentation.	Report generally well-written but may lack clarity or organization. Presentation could be improved.	Report lacks clarity or organization. Presentation needs significant work.	Poorly written report with ineffective or missing presentation.

Criteria	Weight	Maximum marks	Level (1–4)	Marks obtained
Problem Definition & Scope	15%	3.75		
Design & Methodology	20%	5.00		
Implementation & Results	25%	6.25		
Analysis & Interpretation	20%	5.00		
Report & Presentation	20%	5.00		
Total	100%	**25**		

Marks for a criterion = (level awarded ÷ 4) × maximum marks.

Evaluator: _____Signature / Date: _____

ABSTRACT

Machine-learning credit models used by digital micro-lenders can disadvantage women and rural applicants even when gender is excluded, because alternative-data features such as income, UPI activity and device value can act as proxies. This project evaluates an end-to-end pipeline for auditing and mitigating such bias using a synthetic population of 30,000 applicants. In the generator, true repayment ability is designed to be independent of gender and region, while observed features contain structural group effects. A gender-blind logistic-regression model (AUC 0.745) approved 55.6% of men and 40.0% of women, giving a disparate impact ratio (DIR) of 0.718 (95% bootstrap CI 0.682–0.758) and an equal-opportunity difference (EOD) of −0.164. Gender could be predicted from the gender-blind features with AUC 0.897, indicating substantial proxy information. These results describe the synthetic experiment, not a real lender.

Reweighing alone did not reduce the disparity in this experiment (DIR 0.665, compared with 0.718 for the baseline). Group-specific decision thresholds raised DIR to 0.909–0.980 and reduced the absolute equal-opportunity gap from 0.164 to 0.011–0.051. Across ten independently generated datasets, threshold-based methods met the DIR ≥ 0.80 benchmark in all ten runs. Average profit was slightly higher for the equal-opportunity method, but the observed difference was small relative to the variation across runs and should not be interpreted as proof of a statistically significant gain. Under the tested input-noise conditions, the equal-opportunity method retained a higher DIR than the baseline; this does not establish robustness to every possible failure mode. An intersectional gender-by-region audit showed that a residual approval gap remained for rural women. The report provides a technical desk review of the DPDP Act 2023, RBI digital-lending guidance, EU AI Act Annex III and ISO/IEC TR 24027, discusses the legal tension of using gender at decision time, and includes a model card. All results use synthetic data and demonstrate a method, not bias at any actual lender.

Keywords: algorithmic fairness; credit scoring; digital micro-lending; disparate impact; proxy discrimination; reweighing; threshold optimisation; responsible AI.

TABLE OF CONTENTS

Abstract vii

List of Figures x

List of Tables xi

List of Abbreviations xii

Chapter 1: Problem Definition & Operational Relevance 1

1.1 Problem Statement & Background 1

1.2 Industry, Business & Societal Context 1

1.3 Project Objectives 2

1.4 Scope and Operational Boundaries 2

Chapter 2: Theoretical Formulation, Literature Review & Mathematical Framework 4

2.1 Theoretical Foundations 4

2.2 Mathematical and Algorithmic Formulations 5

2.3 State-of-the-Art & Comparative Literature 6

Chapter 3: System Architecture, Design Pipeline & Workflow Models 8

3.1 High-Level Architectural Pipeline 8

3.2 Component-Level Modular Design 9

3.3 Data Flow & Schema Specifications 9

Chapter 4: Implementation Details, Algorithms & Technological Artifacts 11

4.1 Technology Stack & Development Environment 11

4.2 Algorithm Implementation & Pseudocode 11

4.3 Defensive Engineering & Error Handling 12

4.4 User Interface & Interaction Layer 13

Chapter 5: Experimental Evaluation, Testing, Trade-off Analysis & Empirical Results 14

5.1 Test Methodology & Benchmark Datasets 14

5.2 Quantitative Performance Metrics 15

5.3 Comparative Baseline vs. Improved System Analysis 17

5.4 Engineering Trade-off Synthesis 19

5.5 Operational Verification Output 21

Chapter 6: Societal Impact, Ethical Governance, Safety & Security Audits 22

6.1 Vulnerability Assessment & Robustness 22

6.2 Legal & Regulatory Compliance 23

6.3 Societal & Environmental Alignment 24

Chapter 7: Conclusion, Engineering Limitations & Future Enhancements 26

7.1 Summary of Contributions 26

7.2 Technical Limitations 26

7.3 Future Enhancements 27

References 28

Appendix A: Complete Source Code Listing 30

Appendix B: Audit Trace Logs, Model Card & Verification Sheet 36

B.1 Complete audit log 36

B.2 Model card 36

B.3 Verification sheet 37

LIST OF FIGURES

Figure 3.1: High-level architecture of the bias-auditing and mitigation pipeline8

Figure 5.1: Correlation of each feature with gender and region (training split)16

Figure 5.2: Baseline predicted repayment probability by gender (dashed line = approval threshold)17

Figure 5.3: Approval rate by gender for each decision system18

Figure 5.4: Disparate impact ratio (left) and absolute equal-opportunity gap (right)18

Figure 5.5: Profit versus disparate impact ratio (sweep of the female threshold, with all six systems marked)20

Figure 5.6: Console audit scorecard (verbatim output of fair_credit_audit.py)21

LIST OF TABLES

Table 1.1: Project objectives, Bloom level and measurable success criteria2
Table 1.2: Scope summary3
Table 2.1: Group-fairness metrics used in this project6
Table 2.2: Comparison of bias-mitigation approaches considered6
Table 3.1: Component specification9
Table 3.2: Applicant schema and structural effects built into the generator9
Table 4.1: Software stack and versions11
Table 4.2: Verification guardrails built into the pipeline12
Table 5.1: Experimental design14
Table 5.2: Feature means by gender (full synthetic population)14
Table 5.3: Baseline audit scorecard (test set, seed 42)15
Table 5.4: Baseline feature attribution and proxy strength16
Table 5.5: Comparison of baseline and mitigated systems (test set, seed 42)17
Table 5.6: Error-rate and portfolio-quality view (test set, seed 42)19
Table 5.7: Stability across ten independent synthetic datasets (mean ± s.d.)20
Table 5.8: Approval rate by gender × region (test set, seed 42)20
Table 6.1: Noise stress test (test set, seed 42; thresholds frozen)22
Table 6.2: Threat assessment for the lending pipeline22
Table 6.3: Compliance matrix23
Table 6.4: Alignment with UN Sustainable Development Goals24
Table 7.1: Objectives versus outcomes26
Table B.1: Model card for the micro-lending credit-scoring pipeline37
Table B.2: Verification checklist37

LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
AOD	Average Odds Difference
AUC	Area Under the ROC Curve
CIAP	Continuous Internal Assessment for Practical
CO / PO / PSO	Course / Program / Program-Specific Outcome
DIR	Disparate Impact Ratio
DPDP	Digital Personal Data Protection (Act, 2023)
EO	Equal Opportunity
EOD	Equal-Opportunity Difference
FPR / TPR	False / True Positive Rate
GBM	Gradient-Boosted Machine
LR	Logistic Regression
ML	Machine Learning
NBFC	Non-Banking Financial Company
RBI	Reserve Bank of India
SDG	Sustainable Development Goal
SPD	Statistical Parity Difference
UPI	Unified Payments Interface

CHAPTER 1

PROBLEM DEFINITION & OPERATIONAL RELEVANCE

1.1 Problem Statement & Background

Digital micro-lending platforms disburse small-ticket, short-tenure loans through mobile applications, often within minutes and without a branch visit. Because most applicants have thin or no bureau history, these platforms underwrite using *alternative data*: UPI transaction frequency, mobile-recharge regularity, utility-bill punctuality, device characteristics, address stability and social-graph signals. A supervised machine-learning model converts these signals into a repayment probability, and a cut-off score converts that probability into an approve/reject decision.

The technical problem addressed in this project is that such a pipeline can systematically disadvantage protected groups even when the protected attribute is never supplied to the model. Structural inequalities in the real economy (lower recorded income, more informal work, cheaper handsets, shorter address tenure and smaller digital networks for women and rural residents) are encoded into the very features that make the model predictive. The model therefore learns *proxies* for gender and region, and the resulting approval decisions show a disparate impact on groups whose underlying ability to repay is no different. This practice is commonly described as proxy discrimination or algorithmic redlining.

The project designs, implements and evaluates an end-to-end **bias auditing and mitigation pipeline** for a digital micro-lending credit model. It first measures how large the disparity is and where it comes from, then applies pre-processing and post-processing interventions, and finally quantifies what each intervention costs or gains in lending profit.

1.2 Industry, Business & Societal Context

Access to formal credit is a recognised lever for household resilience and enterprise formation, and digital lenders are often the first formal credit source for first-time borrowers. If their scoring models exclude creditworthy women or rural applicants, the

exclusion is both an ethical failure and a commercial one: creditworthy customers are rejected and the lender forgoes interest income that a better-calibrated model would have earned.

Legacy practice falls short in three ways. First, *fairness through unawareness* (deleting the gender column) is the default compliance step, but it does not remove correlated proxies. Second, model validation is usually limited to aggregate accuracy or AUC, which can be excellent while one group is approved far less often. Third, disparity is rarely monitored after deployment, so drift in the applicant mix can quietly widen it.

The consequences of leaving the problem unaddressed are regulatory, economic and reputational. Credit scoring of natural persons is classified as *high-risk* under Annex III of the EU AI Act; India's Digital Personal Data Protection Act, 2023 places accuracy and accountability duties on entities that use personal data for decisions about individuals; and the Reserve Bank of India's digital-lending directions emphasise transparency, consent and grievance redressal. A lender that cannot demonstrate that it has measured and managed bias carries legal, supervisory and brand risk.

1.3 Project Objectives

Four specific, measurable objectives, expressed with Bloom's-taxonomy action verbs, govern the project (Table 1.1).

Table 1.1: Project objectives, Bloom level and measurable success criteria

#	Objective (action verb)	Bloom	Measurable success criterion
O1	**Audit** the baseline gender-blind credit model: quantify disparate impact ratio (DIR), statistical parity difference (SPD), equal-opportunity difference (EOD), average odds difference (AOD) and proxy leakage.	BL4	All metrics reported with 95% bootstrap confidence interval for DIR; proxy-leakage AUC reported.
O2	**Implement** two mitigation families: reweighing (pre-processing) and group-specific decision thresholds (post-processing), plus their combination.	BL3	Three mitigated pipelines running end-to-end from one script.
O3	**Evaluate** the fairness–profit trade-off and its stability across 10 independently generated datasets and under input-noise stress.	BL5	DIR ≥ 0.80 and |TPR gap| driven towards 0.00; profit change per applicant reported.

#	Objective (action verb)	Bloom	Measurable success criterion
O4	**Formulate** a compliance assessment and a model card against DPDP Act 2023, RBI digital-lending directions, EU AI Act Annex III and ISO/IEC TR 24027.	BL5 / BL6	Compliance matrix (Chapter 6) and model card (Appendix B) completed.

1.4 Scope and Operational Boundaries

In scope. A binary approve/reject decision for a single unsecured micro-loan product of ₹10,000; a primary protected attribute of gender (female versus male) and a secondary attribute of region (rural versus urban) used for intersectional analysis; two baseline classifiers (logistic regression and gradient-boosted trees); two mitigation families and their combination; and a business evaluation based on realised portfolio profit.

Data assumption. Real lender data containing gender and repayment outcomes is confidential and, under the DPDP Act 2023, cannot responsibly be collected for a student project. A **synthetic applicant population of 30,000** was therefore generated. Its key design property is that *true repayment depends only on a latent creditworthiness variable that has the same distribution for every group*, while the observed features carry group-dependent structural effects. This gives the audit a known ground truth: any approval gap between groups is, by construction, unjustified. All numerical results in this report are therefore properties of the synthetic generator and must be read as a demonstration of method rather than as an estimate of any real lender's bias.

Technical constraints. The experiments run on a single CPU in Python 3 with scikit-learn 1.8.0; no GPU is needed. Gender is treated as binary; reject inference (learning from applicants who were never funded) and temporal drift are out of scope; the decision threshold is the only operational lever varied in post-processing.

Table 1.2: Scope summary

Dimension	Included	Excluded / assumed
Data	30,000 synthetic applicants, 9 features, 2 protected attributes	Real borrower data; bureau scores; free-text and image data
Models	Logistic regression, histogram gradient boosting	Deep networks; in-processing fairness constraints
Fairness notions	Demographic parity (DIR, SPD), equal opportunity (EOD), average odds (AOD)	Individual fairness; counterfactual fairness

Dimension	Included	Excluded / assumed
Business model	Fixed ₹10,000 loan; 12% net margin; 70% loss given default	Risk-based pricing; variable ticket size
Governance	DPDP, RBI, EU AI Act, ISO/IEC TR 24027 desk review	Formal legal opinion

CHAPTER 2

THEORETICAL FORMULATION, LITERATURE REVIEW & MATHEMATICAL FRAMEWORK

2.1 Theoretical Foundations

2.1.1 Sources of algorithmic bias in credit

Surveys of bias in machine learning distinguish several entry points [1]: *historical bias* (the world already contains inequality), *representation bias* (some groups are under-sampled), *measurement bias* (features or labels are recorded with different accuracy for different groups) and *proxy bias* (neutral variables that are statistically tied to a protected attribute). Digital micro-lending is exposed to all four. The synthetic generator used here concentrates on measurement and proxy bias: income, UPI volume, device value and contact count are all shifted downward for women and rural residents although repayment ability is not.

2.1.2 Fairness definitions

Group-fairness definitions compare the decisions made for different demographic groups [2], [1]. *Demographic parity* requires equal approval rates. *Equal opportunity* [3] requires equal true-positive rates, that is, creditworthy applicants should be approved at the same rate in every group. *Equalized odds* additionally requires equal false-positive rates. *Predictive parity* and *calibration* require that a given score implies the same repayment probability in every group. Individual fairness [4] requires that similar applicants receive similar decisions.

These criteria cannot all hold at once when base rates differ between groups [5], [6], so a lender must choose which to prioritise and justify the choice. Lending adds a legal lens: the *four-fifths rule*, originating in US employment-selection guidelines [7], treats a selection-rate ratio below 0.80 as prima facie evidence of adverse impact. It is used here as a screening benchmark, not as a legal finding.

2.1.3 Mitigation taxonomy

Mitigation techniques act at three stages. *Pre-processing* changes the training data, for example by reweighing instances [8] or repairing feature distributions [9]. *In-processing* adds a fairness term or constraint to the learning objective [10], [11]. *Post-processing* adjusts the outputs of a trained model, for example by choosing group-specific thresholds [3]. The project evaluates a pre-processing method, a post-processing method and their combination because these can be applied to any existing scoring model without retraining its architecture.

2.2 Mathematical and Algorithmic Formulations

Let X denote the feature vector, G ∈ {m, f} the protected attribute, Y ∈ {0, 1} the repayment outcome (1 = repays) and $\hat{Y}$ ∈ {0, 1} the lending decision (1 = approve). The audit metrics are defined in Eqs. (2.1)–(2.4).

```
DIR = P(Yhat=1 | G=f) / P(Yhat=1 | G=m)(2.1)
SPD = P(Yhat=1 | G=f) - P(Yhat=1 | G=m)(2.2)
EOD = P(Yhat=1 | Y=1, G=f) - P(Yhat=1 | Y=1, G=m)(2.3)
AOD = 0.5 * [ (FPR_f - FPR_m) + (TPR_f - TPR_m) ](2.4)
```

Reweighing. Kamiran and Calders [8] assign each training instance a weight that makes the protected attribute and the label statistically independent in the weighted data (Eq. 2.5). Cells in which a group is under-represented among positive outcomes receive weights above one.

```
w(g, y) = P(G=g) * P(Y=y) / P(G=g, Y=y)(2.5)
```

Threshold decision rule. A scoring model returns s(x) = $\hat{P}$(Y=1 | x). The decision rule applies a group-specific cut-off t_g (Eq. 2.6); the baseline sets t_m = t_f.

```
Yhat = 1  if  s(x) >= t_g ,  else 0(2.6)
```

Profit model and break-even threshold. An approved loan that is repaid earns a net margin M = ₹1,200 (12% of ₹10,000); an approved loan that defaults loses L = ₹7,000 (70% loss given default). The expected value is non-negative when $p \times M - (1 - p) \times L \geq 0$. Rearranging gives $p \geq L / (M + L)$ = ₹7,000 / (₹1,200 + ₹7,000) = 0.8537, or approximately 85.4%. This is a threshold implied by the project's assumed loan

economics, not a universal lending threshold. Portfolio profit is calculated as shown in Eq. (2.8).

```
p * M - (1 - p) * L >= 0   =>   p* = L / (M + L) = 7000 / 8200 = 0.854(2.7)
Profit = SUM_i  Yhat_i * [ y_i * M - (1 - y_i) * L ](2.8)
```

Constrained threshold search. For post-processing, thresholds are chosen on a held-out validation set by solving Eq. (2.9) over a grid of 95 candidate values per group. Because profit, approval rate and true-positive rate are additive or separable by group, every (t_m, t_f) pair is evaluated by broadcasting rather than nested loops.

```
max over (t_m, t_f) of Profit(t_m, t_f)  s.t.  |EOD| <= 0.02   (or DIR >=
0.80)(2.9)
```

2.3 State-of-the-Art & Comparative Literature

Empirical studies of fintech credit show that algorithmic underwriting is not automatically fair. Fuster et al. [12] found that machine-learning models in the US mortgage market raised predictive accuracy but distributed the gains unevenly across racial groups. Bartlett et al. [13] observed that algorithmic lenders discriminated less than face-to-face lenders in pricing but did not eliminate disparities. Open-source toolkits such as AI Fairness 360 [14] and Fairlearn [15] now package dozens of the mitigation methods described below, but they leave the choice of the fairness criterion, and the business consequences, to the practitioner.

Table 2.1: Group-fairness metrics used in this project

Metric	Ideal	Interpretation in lending	Reference
Disparate impact ratio (DIR)	1.00 (≥ 0.80 acceptable)	Female approval rate as a fraction of male approval rate	[7]
Statistical parity difference (SPD)	0.00	Absolute gap in approval rates	[2]
Equal-opportunity difference (EOD)	0.00	Gap in approval rate among applicants who would repay	[3]
Average odds difference (AOD)	0.00	Mean of the TPR gap and FPR gap	[14]
Calibration by group	Predicted = actual	Whether a score means the same repayment risk for each group	[5], [6]

Table 2.2: Comparison of bias-mitigation approaches considered

Approach	Stage	Needs gender at decision time?	Strength	Limitation
Fairness through unawareness	None	No	Trivial to implement	Proxies leak the attribute [4]
Reweighing [8]	Pre	No	Model-agnostic; low cost	Only corrects label–group dependence
Disparate-impact remover [9]	Pre	No (training only)	Repairs feature distributions	May discard predictive signal
Fairness constraints [10]	In	No (training only)	Direct control of trade-off	Needs model re-engineering
Adversarial debiasing [11]	In	No (training only)	Removes attribute information	Unstable training; tuning cost
Group thresholds [3]	Post	**Yes**	Optimal for equal opportunity; no retraining	Explicit use of protected attribute

Source: author's synthesis of [3], [8], [9], [14], [15], [10] and [11].

The comparison shows a recurring tension. Methods that avoid using the protected attribute at decision time may be simpler to govern, but their effectiveness depends on whether the mitigation method matches the source of bias. Group-specific post-processing thresholds were effective in this synthetic experiment, but they explicitly use gender at decision time and therefore raise legal and governance questions discussed in Chapter 6.

CHAPTER 3

SYSTEM ARCHITECTURE, DESIGN PIPELINE & WORKFLOW MODELS

3.1 High-Level Architectural Pipeline

The system is organised as a linear audit-then-mitigate pipeline with a governance layer on top (Figure 3.1). Data enters through a synthetic applicant generator, passes through preprocessing, and trains two baseline classifiers from which the protected attribute is deliberately excluded. The *bias audit* and *proxy analysis* modules then examine the baseline decisions; their findings select which *mitigation* is applied; and the mitigated decisions are re-audited and priced by the *business evaluation* module. A governance layer maps the findings to regulatory obligations and produces the model card.

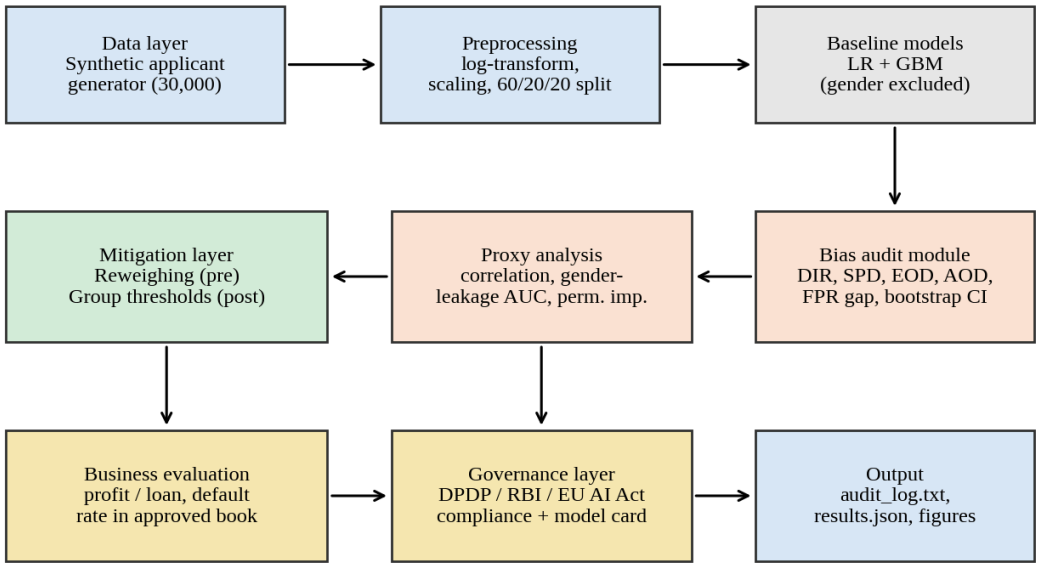


Figure 3.1: High-level architecture of the bias-auditing and mitigation pipeline

Two design decisions are worth stating. First, the protected attribute is held in a separate column that is used only by the audit and mitigation modules and is asserted not to be a

model input. Second, every threshold is selected on a validation split and every reported metric is computed on an untouched test split, so that the reported fairness gains are not artefacts of tuning on the evaluation data.

3.2 Component-Level Modular Design

Table 3.1: Component specification

Component	Input	Output	Interface (function)
Applicant generator	N, random seed	DataFrame of 9 features, gender, region, outcome	generate_applicants()
Preprocessor	DataFrame	Standardised train / validation / test matrices (60/20/20, stratified by gender × outcome)	train_test_split, StandardScaler
Baseline trainer	Train matrix, labels	Scores s(x) from logistic regression and gradient boosting	LogisticRegression, HistGradientBoostingClassifier
Audit engine	Decisions, labels, group vector	DIR, SPD, EOD, AOD, FPR/TPR by group, profit	audit(), dir_bootstrap()
Proxy analyser	Features, gender	Gender-leakage AUC, feature–gender correlations, permutation importance	run_experiment(make_extras=True)
Reweigher	Group vector, labels	Instance weights w(g, y)	reweighing_weights()
Threshold optimiser	Validation scores, labels, groups	Pair (t_m, t_f) maximising profit under a fairness constraint	group_thresholds()
Reporter	All metrics	Console scorecard, audit_log.txt, results.json, six figures	__main__ block, make_figures()

3.3 Data Flow & Schema Specifications

Table 3.2 lists the schema of the synthetic applicant record together with the structural group effect built into each feature. The generating process is summarised in the pseudocode block below; z is a standard-normal latent creditworthiness variable drawn independently of gender and region.

Table 3.2: Applicant schema and structural effects built into the generator

Field	Type	Meaning	Dependence on z	Female effect	Rural effect
age	float	Applicant age (years)	none	none	none
log_income	float	Log of monthly income (₹)	+0.35	−0.30	−0.22
formal_job	0/1	Formally employed	+0.50 (logit)	−1.0 (logit)	−0.8 (logit)
upi_txn	count	UPI transactions per month	+0.30 (log-rate)	−0.30	−0.20
bill_ontime	[0,1]	Share of utility bills paid on time	+0.12	none	none

Field	Type	Meaning	Dependence on z	Female effect	Rural effect
recharge_reg	[0,1]	Mobile-recharge regularity	+0.10	none	none
log_device_value	float	Log of handset value (₹)	+0.10	−0.35	−0.30
addr_stability	months	Months at current address	+4.0	−7 months	none
contacts	count	Contacts on the platform	+0.10 (log-rate)	−0.30	none
female / rural	0/1	Protected / secondary attribute (audit only)	independent	–	–
repaid	0/1	Outcome; P = sigmoid(1.9 + 1.3 z)	–	none	none

```
FOR each applicant i = 1..N:
    female ~ Bernoulli(0.42);  rural ~ Bernoulli(0.50);  z ~ Normal(0, 1)
    features = f(z) + group_shift(female, rural) + noise      # Table 3.2
    repaid   ~ Bernoulli( sigmoid(1.9 + 1.3 * z) )             # no group term
SPLIT 60/20/20 stratified by (female, repaid);  fit scaler on train only
```

The deliberate absence of a gender term in the outcome equation means that the true repayment rates of the two groups are almost identical in the test split (male 81.3%, female 81.8%), so a fair system should approve creditworthy applicants of both groups at similar rates.

CHAPTER 4

IMPLEMENTATION DETAILS, ALGORITHMS & TECHNOLOGICAL ARTIFACTS

4.1 Technology Stack & Development Environment

Table 4.1: Software stack and versions

Layer	Tool	Version	Role
Language / runtime	Python	3.12.3	All pipeline code
Numerics	NumPy / SciPy	2.4.4 / 1.17.1	Vectorised threshold search, statistics
Data handling	pandas	3.0.2	Applicant table and group summaries
Machine learning	scikit-learn [16]	1.8.0	Logistic regression, gradient boosting, splits, permutation importance
Visualisation	Matplotlib	3.10.8	Six publication figures
Hardware	Single CPU core	–	No GPU required; full run including 10-seed robustness takes minutes

Fairness metrics and mitigation methods were implemented directly in NumPy so that every equation in Chapter 2 maps to a visible line of code; the AIF360 [14] and Fairlearn [15] toolkits were not required.

4.2 Algorithm Implementation & Pseudocode

Algorithm 1 gives the pseudocode of the complete audit-and-mitigate procedure. The three functions that carry the core logic (the audit scorecard, reweighing and the constrained group-threshold search) follow, quoted verbatim from the project source (Appendix A).

```
ALGORITHM 1: Audit and mitigate a credit-scoring pipeline
INPUT : applicants D (features X, gender G, outcome Y); loan economics (M, L)
OUTPUT: audit scorecards for baseline and mitigated decision rules
1  Split D into train / validation / test (60/20/20, stratified by G x Y)
2  Fit baseline model on X (gender excluded); get scores s_val, s_test
3  t0 <- argmax_t Profit(s_val >= t)              # single global threshold
4  scorecard_baseline <- AUDIT(s_test >= t0, Y_test, G_test)
5  PROXY analysis: AUC of predicting G from X; corr(X_j, G); permutation importance
6  w <- REWEIGH(G_train, Y_train); refit model with weights w -> scores r_val, r_test
7  (t_m, t_f) <- argmax Profit(val) s.t. |EOD(val)| <= 0.02  (or DIR(val) >= 0.80)
8  decisions <- (score >= t_f) for G = f, (score >= t_m) for G = m
9  scorecard_method <- AUDIT(decisions, Y_test, G_test); bootstrap CI of DIR
10 Repeat steps 1-9 on 10 independent datasets; stress-test with feature noise
```

4.2.1 Audit scorecard

```
def audit(decision, y, g, score=None):
    """Return the full fairness + business scorecard for one decision vector."""
    m, f = (g == 0), (g == 1)
    ar_m, ar_f = decision[m].mean(), decision[f].mean()
```

```
    tpr_m, tpr_f = decision[m & (y == 1)].mean(), decision[f & (y == 1)].mean()
    fpr_m, fpr_f = decision[m & (y == 0)].mean(), decision[f & (y == 0)].mean()
    out = {"approval_male": ar_m, "approval_female": ar_f,
           "DIR": ar_f / ar_m, "SPD": ar_f - ar_m,
           "TPR_male": tpr_m, "TPR_female": tpr_f, "EOD": tpr_f - tpr_m,
           "FPR_male": fpr_m, "FPR_female": fpr_f, "AOD": 0.5 * ((fpr_f - fpr_m) + (tpr_f -
tpr_m)),
           "accuracy": accuracy_score(y, decision), "approved_repay_rate": y[decision ==
1].mean(),
           "profit": profit(decision, y), "profit_per_applicant": profit(decision, y) / len(y),
           "approval_overall": decision.mean()}
    if score is not None:
        out["AUC"] = roc_auc_score(y, score)
    return out
```

4.2.2 Reweighing (pre-processing)

```
def reweighing_weights(g, y):
    """Kamiran & Calders (2012): w(g,y) = P(g) * P(y) / P(g,y)  -> makes g and y independent."""
    w = np.ones(len(y))
    for gv in (0, 1):
        for yv in (0, 1):
            mask = (g == gv) & (y == yv)
            w[mask] = (np.mean(g == gv) * np.mean(y == yv)) / max(mask.mean(), 1e-12)
    return w
```

4.2.3 Constrained group-threshold search (post-processing)

```
def group_thresholds(score, y, g, mode, grid=np.linspace(0.05, 0.99, 95), eo_tol=0.02,
dir_min=0.80):
    """Post-processing: pick (t_male, t_female) maximising validation profit subject to a
fairness
    constraint ('eo' -> |TPR gap| <= eo_tol ; 'dir' -> DIR >= dir_min). Metrics are separable by
group,
    so every pair on the grid is evaluated by broadcasting."""
    m, f = g == 0, g == 1
    pm = np.array([profit(score[m] >= t, y[m]) for t in grid])
    pf = np.array([profit(score[f] >= t, y[f]) for t in grid])
    arm = np.array([(score[m] >= t).mean() for t in grid])
    arf = np.array([(score[f] >= t).mean() for t in grid])
    tm = np.array([(score[m & (y == 1)] >= t).mean() for t in grid])
    tf = np.array([(score[f & (y == 1)] >= t).mean() for t in grid])
    tot = pm[:, None] + pf[None, :]
    if mode == "eo":
        ok = np.abs(tf[None, :] - tm[:, None]) <= eo_tol
    else:
        ok = (arf[None, :] / np.maximum(arm[:, None], 1e-9)) >= dir_min
    tot = np.where(ok, tot, -np.inf)
    if not np.isfinite(tot).any():          # guard: never fall back silently to threshold (0, 0)
        raise ValueError(f"No threshold pair satisfies the '{mode}' fairness constraint on the
validation set")
    i, j = np.unravel_index(np.argmax(tot), tot.shape)
    return float(grid[i]), float(grid[j])
```

4.3 Defensive Engineering & Error Handling

Table 4.2: Verification guardrails built into the pipeline

Risk	Guardrail implemented
Protected attribute leaking into the model	`assert PROTECTED not in FEATURES` executes at import time
Evaluation on tuned data	Thresholds fitted on the validation split only; metrics reported on the test split

Risk	Guardrail implemented
Irreproducible results	Fixed seeds for generation, splitting, boosting, bootstrap and noise; 10 independent seeds for robustness
Division by zero in ratios and weights	Denominators floored (`max(…, 1e-12)`, `np.maximum(…, 1e-9)`)
Silent failure of the constrained search	If no threshold pair is feasible, `group_thresholds` raises `ValueError` instead of defaulting to (0, 0)
Stratification loss	Splits stratified on gender × outcome so every group keeps its outcome mix
Over-claiming from one run	Bootstrap 95% intervals (500 resamples) and mean ± s.d. over 10 datasets

4.4 User Interface & Interaction Layer

The implemented interface is a command-line audit pipeline. Running python fair_credit_audit.py executes the experiment and prints a fixed-width scorecard (reproduced in Section 5.5 and Appendix B). It writes results.json (machine-readable metrics), audit_log.txt (the console report), and a figs/ directory of PNG charts. This command-line output is the implemented interface in this version.

A graphical dashboard in Streamlit or Gradio, with a threshold slider and live updates to DIR and profit, is not implemented in this version and remains future work. Figure 5.5 provides a static view of the trade-off and is not an interactive dashboard.

CHAPTER 5

EXPERIMENTAL EVALUATION, TESTING, TRADE-OFF ANALYSIS & EMPIRICAL RESULTS

5.1 Test Methodology & Benchmark Datasets

Evaluation uses the synthetic population described in Section 1.4. The 30,000 applicants were split 60/20/20 into 18,000 training, 6,000 validation and 6,000 test records, stratified by gender and outcome. Women form 42.3% of the test set. All headline numbers below come from the test split of the primary run (seed 42); robustness is then assessed over ten further datasets (seeds 1–10). Because the data are synthetic, there is no external benchmark dataset, and the generator itself is the benchmark: it guarantees equal true repayment ability across groups.

Table 5.1: Experimental design

Element	Specification
Population / splits	30,000 synthetic applicants; 18,000 / 6,000 / 6,000 (train / validation / test)
Baselines	B1: logistic regression (L2); B2: histogram gradient boosting (200 iterations, learning rate 0.08). Gender excluded from inputs; one global profit-optimal threshold.
Mitigations	M1: reweighing then global threshold; M2: equal-opportunity group thresholds ($\|\text{TPR gap}\| \leq 0.02$); M3: group thresholds enforcing DIR ≥ 0.80; M4: reweighing + equal-opportunity thresholds
Protected attributes	Gender (primary audit); gender × region (intersectional audit)
Economics	Loan ₹10,000; margin ₹1,200 if repaid; loss ₹7,000 if default; break-even repayment probability 0.854
Uncertainty	Bootstrap 95% CI for DIR (500 resamples); 10 independent datasets; input-noise stress test

Table 5.2 confirms that the generator behaves as designed: women and rural applicants have markedly lower recorded income, UPI activity, device value, address tenure and platform contacts, while the behavioural signals (bill punctuality, recharge regularity) and the true repayment rate are the same across groups.

Table 5.2: Feature means by gender (full synthetic population)

Feature	Male mean	Female mean	Difference
log monthly income	9.59	9.29	−0.30

Feature	Male mean	Female mean	Difference
formally employed (share)	0.32	0.16	−0.16
UPI transactions / month	14.1	10.4	−3.7
log device value	9.25	8.90	−0.35
address stability (months)	27.1	20.2	−6.8
platform contacts	33.3	24.7	−8.6
bills paid on time (share)	0.66	0.66	+0.00
recharge regularity	0.60	0.60	−0.00
true repayment rate	0.81	0.82	+0.01

5.2 Quantitative Performance Metrics

5.2.1 Baseline audit

The gender-blind baseline logistic regression achieves a respectable AUC of 0.745, yet it approves 55.6% of men and only 40.0% of women. The disparate impact ratio is **0.718** (95% bootstrap CI 0.682–0.758), below the 0.80 screening benchmark with the entire interval below it. Among applicants who would in fact repay, 62.0% of men but only 45.6% of women are approved: an equal-opportunity gap of −0.164. The gradient-boosted comparator shows the same pattern (DIR 0.758, EOD −0.146), so the disparity is not an artefact of the linear model.

Table 5.3: Baseline audit scorecard (test set, seed 42)

Metric	Baseline LR	Baseline GBM
AUC	0.745	0.734
Approval rate: male / female	55.6% / 40.0%	57.1% / 43.3%
Disparate impact ratio (DIR)	0.718	0.758
Statistical parity difference	−0.157	−0.138
TPR: male / female	62.0% / 45.6%	63.3% / 48.7%
Equal-opportunity difference (EOD)	−0.164	−0.146
FPR: male / female	28.2% / 14.5%	30.2% / 18.7%
Average odds difference (AOD)	−0.150	−0.131
Repayment rate of approved loans	91.5%	90.8%
Profit per applicant	₹246	₹230

5.2.2 Where the bias comes from

Although gender is excluded, a separate classifier can recover it from the remaining features with an AUC of **0.897**, confirming heavy proxy leakage. Figure 5.1 shows the correlation of each feature with gender and region; contacts, device value, UPI activity and income are the strongest carriers. The permutation-importance analysis in Table 5.4, however, shows that the baseline relies mainly on bill punctuality (AUC drop 0.0578), income (0.0243), UPI activity (0.0195) and recharge regularity (0.0161). The harm therefore flows mainly through **income and UPI volume**: variables that are legitimately predictive of repayment but are measured lower for women. Pure proxies such as contacts and device value are largely ignored by the model once better signals are available.

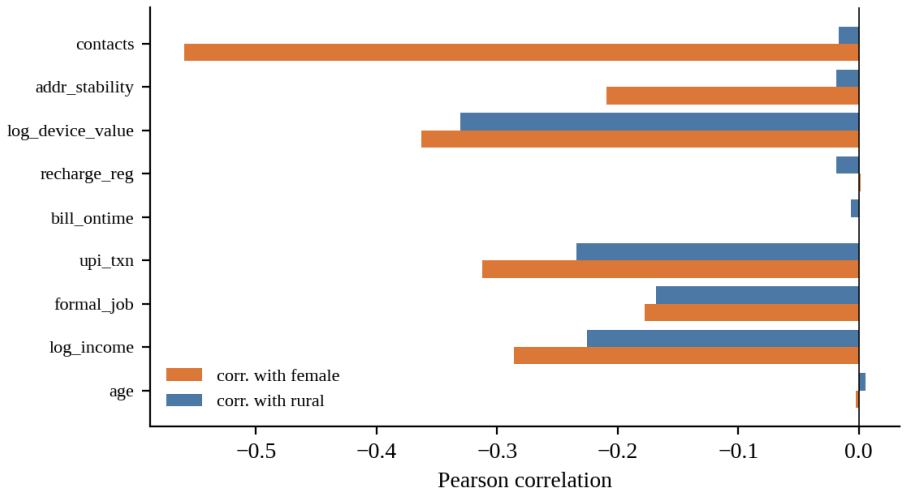


Figure 5.1: Correlation of each feature with gender and region (training split)

Table 5.4: Baseline feature attribution and proxy strength

Feature	Permutation importance (ΔAUC)	LR coefficient (standardised)	Correlation with female
bill_ontime	0.0578	+0.500	−0.000
log_income	0.0243	+0.344	−0.286
upi_txn	0.0195	+0.301	−0.312
recharge_reg	0.0161	+0.299	+0.002
log_device_value	0.0012	−0.078	−0.363
addr_stability	0.0004	+0.036	−0.209
contacts	0.0002	+0.042	−0.560
age	-0.0001	−0.034	−0.003
formal_job	-0.0001	+0.007	−0.178

The calibration check explains the mechanism. The baseline predicts a mean repayment probability of 0.834 for men against an actual rate of 0.813 (over-prediction), and 0.786 for women against an actual rate of 0.818 (under-prediction). The model is *miscalibrated against women*: it systematically understates their creditworthiness because their income and transaction footprints are depressed. Figure 5.2 shows the resulting leftward shift of the female score distribution relative to the single approval threshold.

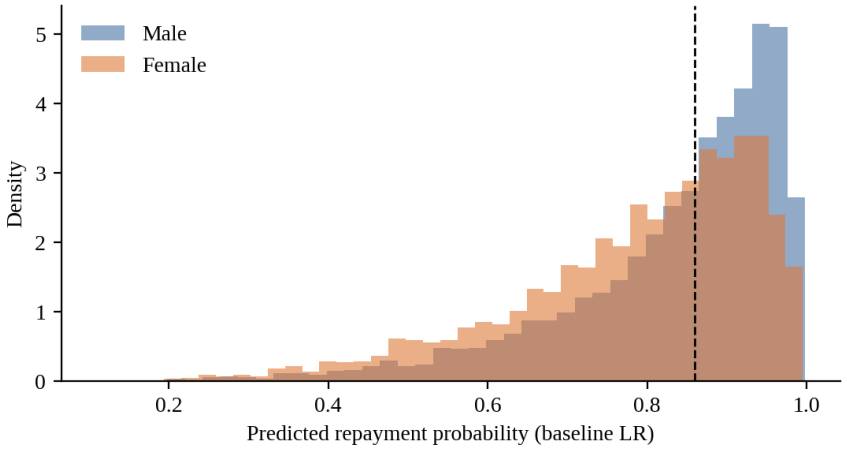


Figure 5.2: Baseline predicted repayment probability by gender (dashed line = approval threshold)

5.3 Comparative Baseline vs. Improved System Analysis

Table 5.5 compares all six decision systems on the same test set, and Figures 5.3 and 5.4 visualise the approval rates, the disparate impact ratio and the equal-opportunity gap.

Table 5.5: Comparison of baseline and mitigated systems (test set, seed 42)

System	Approval M	Approval F	DIR	SPD	EOD	AUC	Profit / applicant
Baseline LR	55.6%	40.0%	0.718	−0.157	−0.164	0.745	₹246
Baseline GBM	57.1%	43.3%	0.758	−0.138	−0.146	0.734	₹230
Reweighing	49.5%	32.9%	0.665	−0.166	−0.181	0.745	₹253
EO thresholds	52.9%	48.1%	0.909	−0.048	−0.051	0.745	₹268
DIR ≥ 0.80 thresholds	49.1%	48.1%	0.980	−0.010	−0.011	0.745	₹267
Reweigh + EO	49.5%	47.9%	0.967	−0.016	−0.020	0.745	₹269

Thresholds (male / female): baseline 0.86 / 0.86; EO 0.87 / 0.83; DIR ≥ 0.80 0.88 / 0.83; Reweigh + EO 0.88 / 0.83. DIR 95% CI: baseline 0.682–0.758; EO 0.870–0.959; DIR ≥ 0.80 0.932–1.034; Reweigh + EO 0.922–1.021.

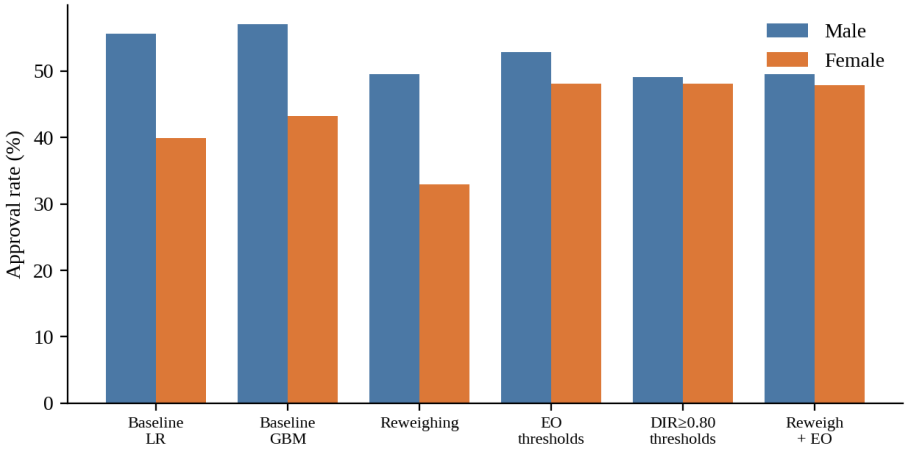


Figure 5.3: Approval rate by gender for each decision system

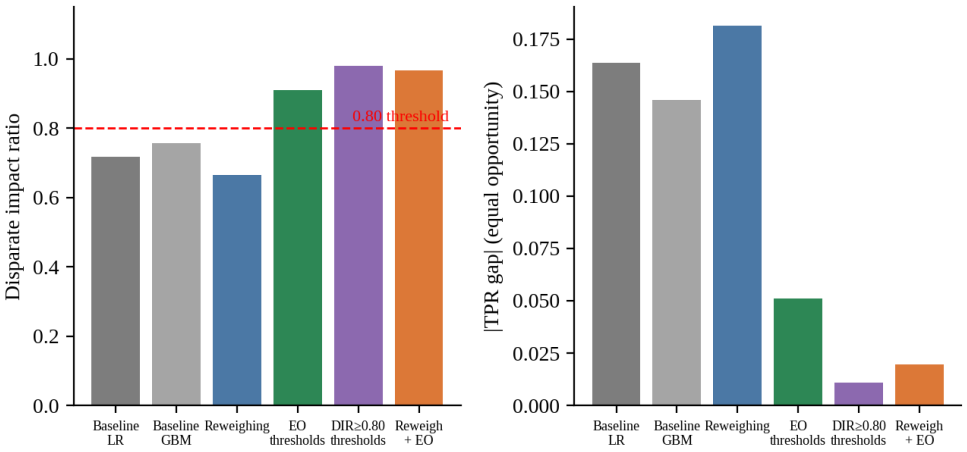


Figure 5.4: Disparate impact ratio (left) and absolute equal-opportunity gap (right)

5.3.1 Reweighing: a negative result that explains the mechanism

Reweighing leaves the disparity essentially unchanged (DIR 0.665 versus 0.718; over ten datasets 0.740 versus 0.738). The reason is visible in the data: the weights span only 0.987–1.018 because gender and the repayment label are already almost independent (Table 5.2). Reweighing corrects *label-level* dependence between group and outcome, whereas the bias here is a *feature-level* measurement shift. This is an instructive finding: a mitigation must match the bias mechanism, and the audit must come first.

5.3.2 Group thresholds: effective and, here, profit-neutral

Group-specific thresholds raise the female approval rate from 40.0% to 48.1% while the male rate moves from 55.6% to 52.9%. The equal-opportunity variant lifts DIR to **0.909**

(CI 0.870–0.959) and cuts the TPR gap from −0.164 to −0.051. The DIR-targeted variant reaches 0.980 and an EOD of −0.011; the profit-maximiser lands near parity because, with a miscalibrated model, approving more creditworthy women is the profitable move. The combination of reweighing and equal-opportunity thresholds gives DIR 0.967 and EOD −0.020. Neither threshold method changes the scores, so AUC stays at 0.745.

The validation-tuned equal-opportunity constraint ($|EOD| \leq 0.02$) is not met exactly on the test split of seed 42 (test EOD −0.051), a small generalisation gap caused by sampling noise in the validation TPRs; over ten datasets the mean EOD is -0.010 ± 0.021.

Table 5.6: Error-rate and portfolio-quality view (test set, seed 42)

System	FPR M	FPR F	AOD	Accuracy	Repay rate of approved loans
Baseline LR	28.2%	14.5%	−0.150	59.2%	91.5%
Baseline GBM	30.2%	18.7%	−0.131	60.4%	90.8%
Reweighing	21.4%	10.6%	−0.145	54.8%	92.6%
EO thresholds	24.5%	20.0%	−0.048	61.0%	91.8%
DIR ≥ 0.80 thresholds	21.6%	20.0%	−0.014	59.5%	92.1%
Reweigh + EO	21.4%	20.2%	−0.016	59.6%	92.1%

Mitigation does not degrade portfolio quality: the repayment rate of approved loans stays between 91.5% and 92.1%. The female false-positive rate rises (14.5% to 20.0% under EO thresholds), which is the expected price of approving more women: the model's errors are redistributed rather than removed.

5.4 Engineering Trade-off Synthesis

5.4.1 Fairness versus profit

Figure 5.5 sweeps the female threshold while holding the male threshold at its baseline value. Profit is not monotonic in DIR: it rises from ₹14,78,000 at the baseline to a peak of ₹15,59,600 near DIR = 1.00, then falls as the threshold is relaxed too far, and drops below the baseline profit once DIR exceeds roughly 1.16. Over-correction therefore carries a real cost: approving women who fall below the break-even repayment probability destroys value. In this setting fairness and profit are aligned up to parity because the baseline *under*-serves a profitable segment; they conflict only beyond it.

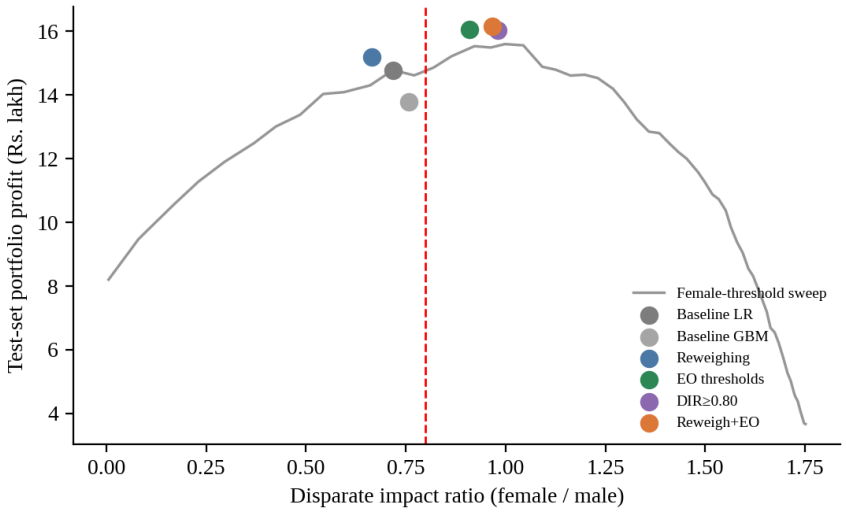


Figure 5.5: Profit versus disparate impact ratio (sweep of the female threshold, with all six systems marked)

5.4.2 Robustness over ten datasets

Table 5.7: Stability across ten independent synthetic datasets (mean ± s.d.)

System	DIR	Min DIR	Seeds with DIR ≥ 0.80	EOD	Profit / applicant (₹)
Baseline LR	0.738 ± 0.030	0.670	0 / 10	−0.159 ± 0.015	276 ± 14
Reweighing	0.740 ± 0.031	0.666	0 / 10	−0.158 ± 0.017	275 ± 13
EO thresholds	0.984 ± 0.040	0.931	10 / 10	−0.010 ± 0.021	282 ± 17
DIR ≥ 0.80 thresholds	0.937 ± 0.082	0.834	10 / 10	−0.036 ± 0.047	281 ± 14
Reweigh + EO	0.990 ± 0.035	0.923	10 / 10	−0.007 ± 0.020	282 ± 18

The baseline fails the 0.80 benchmark in all ten datasets (minimum DIR 0.670); both threshold-based mitigation methods meet the benchmark in all ten runs. Across datasets, mean profit per applicant was ₹276 for the baseline and ₹282 for the equal-opportunity system, while the standard deviations were ₹14 and ₹17 respectively. This descriptive comparison does not establish a statistically significant profit gain or penalty; a formal paired statistical test would be needed to support such a claim.

5.4.3 Intersectional effects

Table 5.8: Approval rate by gender × region (test set, seed 42)

Group	Baseline LR	EO thresholds	Reweigh + EO
M Urban	58.7%	55.9%	52.7%
M Rural	52.5%	49.8%	46.3%

Group	Baseline LR	EO thresholds	Reweigh + EO
F Urban	42.8%	51.7%	51.7%
F Rural	37.1%	44.6%	44.2%
Ratio: F-Rural / M-Urban	0.632	0.798	0.839

Gender-only mitigation leaves a residual intersectional gap. Rural women are the worst-served group at baseline (approval 37.1% against 58.7% for urban men, a ratio of 0.632). After equal-opportunity thresholds the ratio is 0.798, marginally under the 0.80 benchmark, because region was never targeted. Fairness audits that look at a single attribute can therefore certify a system that still disadvantages a compound group.

5.4.4 Summary of trade-offs observed

- **Fairness versus accuracy:** post-processing leaves AUC unchanged by construction; decision-level accuracy moves only from 59.2% to 61.0%, and accuracy is a poor objective for a lender whose outcomes are asymmetric.
- **Fairness versus profit:** neutral up to parity, costly beyond it (Figure 5.5).
- **Fairness versus legal simplicity:** the most effective method uses gender at decision time (Chapter 6).
- **Criterion versus criterion:** equal opportunity and demographic parity are met together here only because base rates are nearly equal; with unequal base rates they conflict [5], [6].

5.5 Operational Verification Output

The pipeline was executed end-to-end from the command line. Figure 5.6 reproduces the console scorecard verbatim from `audit_log.txt`; the full log is given in Appendix B. The student should additionally insert a screenshot of their own terminal run at this point, as required by the report guidelines.

```
================================================================================
FAIR CREDIT AUDIT - digital micro-lending (synthetic data, seed 42)
================================================================================
Applicants: 30000  | split train/val/test = 18000/6000/6000
Female share (test): 0.423 | true repay rate  M: 0.813  F: 0.818
Proxy leakage: gender recoverable from gender-blind features, AUC = 0.897
--------------------------------------------------------------------------------
Method                  AR_M   AR_F    DIR     SPD  TPR_M  TPR_F     EOD    AUC  Profit/app
Baseline LR            0.556  0.400  0.718  -0.157  0.620  0.456  -0.164  0.745       246.3
Baseline GBM           0.571  0.433  0.758  -0.138  0.633  0.487  -0.146  0.734       229.6
Reweighing             0.495  0.329  0.665  -0.166  0.560  0.379  -0.181  0.745       253.3
```

```
EO thresholds           0.529  0.481  0.909  -0.048  0.595  0.544  -0.051  0.745       267.6
DIR>=0.80 thresholds    0.491  0.481  0.980  -0.010  0.555  0.544  -0.011  0.745       267.1
Reweigh + EO            0.495  0.479  0.967  -0.016  0.560  0.541  -0.020  0.745       269.1
--------------------------------------------------------------------------------
```

Figure 5.6: Console audit scorecard (verbatim output of fair_credit_audit.py)

CHAPTER 6

SOCIETAL IMPACT, ETHICAL GOVERNANCE, SAFETY & SECURITY AUDITS

6.1 Vulnerability Assessment & Robustness

Robustness was examined empirically for one failure mode and qualitatively for the others. **Empirical test (input noise).** Gaussian noise of standard deviation σ (in standardised feature units) was added to every test-set feature, simulating mis-recorded transactions or noisy device telemetry, while the model and thresholds were kept frozen. Table 6.1 shows that the fairness benefit of the equal-opportunity thresholds persists as the data degrade: DIR stays between 0.909 and 0.932 against a baseline of roughly 0.72–0.76, while AUC falls from 0.745 to 0.701 at σ = 1.0. Profit per applicant falls sharply at σ = 1.0 for both systems, which shows that the dominant risk under heavy noise is commercial rather than fairness-related.

Table 6.1: Noise stress test (test set, seed 42; thresholds frozen)

σ	AUC	DIR baseline	DIR EO	EOD baseline	EOD EO	Profit/app baseline	Profit/app EO
0.00	0.745	0.718	0.909	−0.164	−0.051	₹246	₹268
0.25	0.738	0.730	0.915	−0.158	−0.045	₹245	₹253
0.50	0.732	0.716	0.918	−0.166	−0.047	₹251	₹250
1.00	0.701	0.758	0.932	−0.141	−0.035	₹179	₹187

Qualitative threat assessment. The remaining threats were assessed by desk analysis only and have **not** been tested experimentally in this project (Table 6.2).

Table 6.2: Threat assessment for the lending pipeline

Threat	Description	Status / control
Feature gaming	Applicants inflate UPI activity, recharge frequency or platform contacts to raise their score	Not tested. Prefer features that are costly to fake; monitor feature drift
Data poisoning	Manipulated repayment records enter retraining data	Not tested. Provenance checks and retraining review gates
Attribute inference	Gender is recoverable from the gender-blind features (AUC 0.897), so the model's inputs are effectively sensitive data	Measured. Treat feature store as sensitive personal data under DPDP; restrict access
Threshold tampering	Insider edits group thresholds to favour or disfavour a group	Not tested. Version-control thresholds; dual approval; audit log

Threat	Description	Status / control
Distribution drift	Applicant mix or economy changes and fairness decays silently	Partly tested via noise and 10 datasets; add scheduled DIR monitoring
Membership / model inversion	Attacker infers who was in training data	Not tested; out of scope for tabular synthetic data
Adversarial evasion	Crafted inputs to cross the approval threshold	Not tested; noise test is a coarse proxy only

6.2 Legal & Regulatory Compliance

Table 6.3 audits the system and this project against the instruments most relevant to an Indian digital lender with potential European exposure. It is a technical desk review, **not legal advice**; regulatory timelines in particular have been moving and should be re-verified before any deployment.

Table 6.3: Compliance matrix

Instrument	Relevant requirement	Project evidence	Residual gap
DPDP Act 2023 [17]	Lawful, consented, purpose-limited processing; duty to ensure accuracy of data used in decisions affecting the individual; security safeguards; breach notification; grievance redressal; additional duties for significant data fiduciaries (e.g., impact assessments, audits)	Synthetic data only, so no personal data processed; accuracy and calibration audited by group; proxy leakage shows features are effectively sensitive	Consent flows, retention schedule and DPIA for a live deployment; confirm the status of the implementing rules
RBI digital-lending guidelines [18]; RBI FREE-AI report [19]	Transparency to borrowers, need-based data collection with explicit consent, board-approved policies, grievance mechanism; FREE-AI sets out fairness, accountability and oversight principles for AI in finance	Audit metrics, model card and human-oversight protocol provide the evidence a board policy would need	These documents do not prescribe a numeric fairness metric; the lender must set and justify its own tolerance
EU AI Act, Annex III item 5(b) [20]	AI systems that evaluate creditworthiness of natural persons are high-risk: risk management, data governance incl. bias examination, technical documentation, logging, transparency, human oversight, accuracy and robustness	Bias examination (Ch. 5), robustness test (6.1), model card (App. B), logging via audit_log.txt	Conformity assessment, post-market monitoring; verify the application timeline, which has been subject to postponement proposals
ISO/IEC TR 24027 [21]; ISO/IEC 42001 [22]	Guidance on bias sources, metrics and assessment; management-system controls for AI	Metrics and bias-source taxonomy follow TR 24027	No management-system certification scope
Model-card practice [23]	Document intended use, metrics by group, limitations	Appendix B model card	Must be maintained with each model version

6.2.1 A legal tension in the best-performing mitigation

The most effective mitigation in this project, group-specific thresholds, uses gender **at decision time**. Many credit-equality regimes prohibit considering a protected characteristic when deciding an individual application: for example the US Equal Credit Opportunity Act and Regulation B [24], and, in the EU, the gender-equal-treatment directive for goods and services [25] as applied by the Court of Justice in *Test-Achats* [26]. Whether a corrective, fairness-motivated use is permissible varies by jurisdiction and is contested. The scientifically cautious reading of this project's results is therefore:

- use group-aware thresholds as an **audit and shadow-pricing tool** to measure how large the disparity is and how much room exists to close it;
- prefer mitigations that need no protected attribute at inference (pre-processing, in-processing constraints [10], feature redesign) for production, and test whether they reach the same DIR;
- obtain legal advice and document business necessity before any production use of group-specific cut-offs.

The reweighing experiment shows why the second route is not automatic: in this data it did nothing, because the bias was in the features, not the labels. Feature redesign (for example, replacing raw income with income *stability* or cash-flow regularity) is the route most likely to work, and testing it is listed as future work.

Finally, the four-fifths rule used as a benchmark is a screening heuristic from employment law [7]; satisfying it is neither necessary nor sufficient for legal compliance in credit.

6.3 Societal & Environmental Alignment

The project's contribution to the United Nations Sustainable Development Goals [27] is summarised in Table 6.4.

Table 6.4: Alignment with UN Sustainable Development Goals

SDG	Relevant target	Contribution of this project
SDG 5: Gender equality	5.a: equal rights to economic resources and access to financial services	Measures and reduces the approval gap for women: DIR 0.718 → 0.909

SDG	Relevant target	Contribution of this project
SDG 10: Reduced inequalities	10.3: ensure equal opportunity and reduce inequalities of outcome	Intersectional audit exposes the rural-women gap that gender-only fixes miss
SDG 8: Decent work and economic growth	8.10: expand access to financial services for all	Recovers creditworthy borrowers wrongly rejected, at no significant profit cost
SDG 1: No poverty	1.4: equal access to economic resources and financial services, including microfinance	Fairer first-time credit access for thin-file applicants
SDG 9: Industry, innovation, infrastructure	9.3: access to financial services for small enterprises	Responsible AI practice for fintech underwriting

Human oversight protocol. A responsible deployment of this pipeline would include: (i) a monthly fairness report of DIR and EOD by gender, region and their intersections, with an alert when DIR falls below 0.85 (a margin above the 0.80 benchmark); (ii) mandatory human review of applications within a band around the approval threshold; (iii) adverse-action reasons stated in plain language and logged; (iv) an appeal route to a grievance officer; and (v) rollback of any model or threshold version that breaches the alert level.

Environmental footprint. The models are small (logistic regression and shallow boosting) and train on a CPU in seconds; the entire experiment, including ten repeated datasets, runs in minutes. The environmental cost of the audit is negligible compared with its potential to prevent wasteful, mis-targeted lending.

CHAPTER 7

CONCLUSION, ENGINEERING LIMITATIONS & FUTURE ENHANCEMENTS

7.1 Summary of Contributions

The project built a reproducible pipeline that audits a gender-blind digital micro-lending model, locates the source of its bias, applies mitigations and prices them. Table 7.1 compares the deliverables with the objectives set in Chapter 1.

Table 7.1: Objectives versus outcomes

Objective	Outcome	Evidence
O1 Audit baseline	Baseline DIR 0.718 (CI 0.682–0.758); EOD −0.164; gender recoverable from features with AUC 0.897; model under-predicts women's repayment	Tables 5.3, 5.4; Figures 5.1–5.2
O2 Implement mitigations	Reweighing, equal-opportunity thresholds, DIR-targeted thresholds and a combined pipeline implemented; reweighing found ineffective for feature-level bias	Sections 4.2, 5.3
O3 Evaluate trade-offs	DIR rises to 0.909–0.980; |EOD| falls to 0.020–0.051; passes 0.80 in 10/10 datasets; profit neutral within noise; over-correction costly	Tables 5.5–5.8; Figure 5.5; Table 6.1
O4 Compliance and model card	Compliance matrix completed; legal tension around group thresholds identified; model card provided	Table 6.3; Appendix B

Three conclusions stand out. First, removing the protected attribute is not enough: proxies carry it back in. Second, the right mitigation depends on the bias mechanism; here, post-processing worked and reweighing did not. Third, in a lending setting where the baseline model under-serves a creditworthy segment, closing the fairness gap up to parity costs little or nothing, whereas over-correcting does cost value.

7.2 Technical Limitations

- **Synthetic data.** Every number is a property of the generator. The magnitude of bias, the profit effect and even the reweighing result may differ on real data; the project validates a method, not a real lender.

- **Selective labels.** Real lenders observe repayment only for approved applicants; reject inference was not modelled.
- **Binary, single-attribute framing.** Gender was treated as binary, and mitigation targeted gender only; the intersectional audit showed the resulting residual gap.
- **Validation-to-test shift.** Thresholds met the 0.02 EOD constraint on validation but not exactly on test; the DIR-targeted variant showed the widest spread across seeds.
- **Simplified economics.** A single ticket size, fixed margin and loss rate; no risk-based pricing, funding cost or operating cost.
- **Untested threats.** Privacy attacks and adversarial evasion were assessed qualitatively only; no graphical dashboard or screenshots from a deployed interface exist.
- **Legal status of group thresholds** is unresolved and jurisdiction-specific (Section 6.2.1).

7.3 Future Enhancements

1. **Real, consented data.** Repeat the audit on anonymised lender data through a regulatory sandbox or a partner NBFC, with protected attributes collected under DPDP-compliant consent or imputed under strict controls.
2. **Attribute-free mitigation.** Implement in-processing fairness constraints [10] and adversarial debiasing [11], and redesign features (stability and cash-flow regularity) to test whether DIR ≥ 0.80 is achievable without using gender at decision time.
3. **Intersectional and calibration constraints.** Extend the threshold search to gender $\times$ region cells and add within-group calibration as a constraint.
4. **Reject inference and drift monitoring.** Model selective labels and add scheduled fairness monitoring with alerts.
5. **Interactive dashboard and explanations.** Build the Streamlit or Gradio interface (threshold slider, live DIR and profit) and SHAP-based adverse-action explanations for borrowers.
6. **Security testing.** Add membership-inference, feature-gaming and evasion experiments and differential-privacy training to the evaluation.

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APPENDIX A

COMPLETE SOURCE CODE LISTING

File: fair_credit_audit.py (reproduces every number, table and figure in Chapters 5 and 6). Run with Python 3 and the packages in Table 4.1.

```
"""
Fair Credit: Algorithmic Bias Auditing and Mitigation in Digital Micro-Lending
Trustworthy & Responsible AI (ADCOR1PC303) - Technical Micro-Project

Pipeline: synthetic applicant generator -> baseline credit model -> fairness audit
          -> mitigation (reweighing, group thresholds) -> profit/fairness trade-off.
Run:      python fair_credit_audit.py        (writes results.json, audit_log.txt, figs/)
"""
import json, sys
import numpy as np
import pandas as pd
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import HistGradientBoostingClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import roc_auc_score, accuracy_score
from sklearn.inspection import permutation_importance

# ---------------------------------------------------------------- configuration
N = 30000                    # synthetic applicants
LOAN = 10_000                # INR principal per loan
GAIN = 0.12 * LOAN           # net interest income when repaid
LOSS = 0.70 * LOAN           # loss given default (30 % recovery)
FEATURES = ["age", "log_income", "formal_job", "upi_txn", "bill_ontime",
            "recharge_reg", "log_device_value", "addr_stability", "contacts"]
PROTECTED = "female"         # used ONLY for auditing / mitigation, never as a model input
plt.rcParams.update({"font.family": "serif", "font.serif": ["Liberation Serif", "Times New Roman"],
                     "font.size": 10, "axes.spines.top": False, "axes.spines.right": False})


assert PROTECTED not in FEATURES, "protected attribute must never be a model input"


def sigmoid(x):
    return 1.0 / (1.0 + np.exp(-x))


# ---------------------------------------------------------------- data generation
def generate_applicants(n=N, seed=42):
    """Synthetic digital-lending applicants.
    True repayment depends ONLY on latent ability z (same distribution for all groups).
    Observed features carry structural group effects (income gap, informal work,
    cheaper devices, fewer contacts), so they act as proxies for gender / region."""
    r = np.random.default_rng(seed)
    female = r.binomial(1, 0.42, n)
    rural = r.binomial(1, 0.50, n)
    z = r.normal(0, 1, n)
    age = np.clip(r.normal(34, 9, n), 18, 65)
    income = np.exp(9.7 + 0.35 * z - 0.30 * female - 0.22 * rural + r.normal(0, 0.35, n))
    formal = r.binomial(1, sigmoid(-0.4 + 0.5 * z - 1.0 * female - 0.8 * rural))
    upi = r.poisson(np.exp(2.7 + 0.30 * z - 0.30 * female - 0.20 * rural))
    bill = np.clip(0.66 + 0.12 * z + r.normal(0, 0.12, n), 0, 1)
    recharge = np.clip(0.60 + 0.10 * z + r.normal(0, 0.14, n), 0, 1)
    device = np.exp(9.4 + 0.10 * z - 0.35 * female - 0.30 * rural + r.normal(0, 0.40, n))
    addr = np.clip(r.gamma(3.0, 9.0, n) - 7.0 * female + 4.0 * z, 1, None)
    contacts = r.poisson(np.exp(3.5 + 0.10 * z - 0.30 * female))
    repaid = r.binomial(1, sigmoid(1.9 + 1.3 * z))            # 1 = repays, 0 = defaults
    return pd.DataFrame({"age": age, "log_income": np.log(income), "formal_job": formal,
                         "upi_txn": upi, "bill_ontime": bill, "recharge_reg": recharge,
                         "log_device_value": np.log(device), "addr_stability": addr,
```

```
                        "contacts": contacts, "female": female, "rural": rural, "repaid": repaid})


# ------------------------------------------------------------------ fairness metrics
def profit(decision, y):
    """Portfolio profit: approved & repaid -> +GAIN ; approved & default -> -LOSS."""
    return float(np.sum(decision * (y * GAIN - (1 - y) * LOSS)))


def audit(decision, y, g, score=None):
    """Return the full fairness + business scorecard for one decision vector."""
    m, f = (g == 0), (g == 1)
    ar_m, ar_f = decision[m].mean(), decision[f].mean()
    tpr_m, tpr_f = decision[m & (y == 1)].mean(), decision[f & (y == 1)].mean()
    fpr_m, fpr_f = decision[m & (y == 0)].mean(), decision[f & (y == 0)].mean()
    out = {"approval_male": ar_m, "approval_female": ar_f,
           "DIR": ar_f / ar_m, "SPD": ar_f - ar_m,
           "TPR_male": tpr_m, "TPR_female": tpr_f, "EOD": tpr_f - tpr_m,
           "FPR_male": fpr_m, "FPR_female": fpr_f, "AOD": 0.5 * ((fpr_f - fpr_m) + (tpr_f - tpr_m)),
           "accuracy": accuracy_score(y, decision), "approved_repay_rate": y[decision == 1].mean(),
           "profit": profit(decision, y), "profit_per_applicant": profit(decision, y) / len(y),
           "approval_overall": decision.mean()}
    if score is not None:
        out["AUC"] = roc_auc_score(y, score)
    return out


def dir_bootstrap(decision, g, n_boot=500, seed=0):
    r = np.random.default_rng(seed)
    vals = []
    for _ in range(n_boot):
        i = r.integers(0, len(g), len(g))
        d, gg = decision[i], g[i]
        vals.append(d[gg == 1].mean() / d[gg == 0].mean())
    return float(np.percentile(vals, 2.5)), float(np.percentile(vals, 97.5))


# ------------------------------------------------------------------ mitigation
def reweighing_weights(g, y):
    """Kamiran & Calders (2012): w(g,y) = P(g) * P(y) / P(g,y)  -> makes g and y independent."""
    w = np.ones(len(y))
    for gv in (0, 1):
        for yv in (0, 1):
            mask = (g == gv) & (y == yv)
            w[mask] = (np.mean(g == gv) * np.mean(y == yv)) / max(mask.mean(), 1e-12)
    return w


def best_global_threshold(score, y, grid=np.linspace(0.05, 0.99, 95)):
    pr = [profit(score >= t, y) for t in grid]
    return float(grid[int(np.argmax(pr))])


def group_thresholds(score, y, g, mode, grid=np.linspace(0.05, 0.99, 95), eo_tol=0.02, dir_min=0.80):
    """Post-processing: pick (t_male, t_female) maximising validation profit subject to a fairness
    constraint ('eo' -> |TPR gap| <= eo_tol ; 'dir' -> DIR >= dir_min). Metrics are separable by group,
    so every pair on the grid is evaluated by broadcasting."""
    m, f = g == 0, g == 1
    pm = np.array([profit(score[m] >= t, y[m]) for t in grid])
    pf = np.array([profit(score[f] >= t, y[f]) for t in grid])
    arm = np.array([(score[m] >= t).mean() for t in grid])
    arf = np.array([(score[f] >= t).mean() for t in grid])
    tm = np.array([(score[m & (y == 1)] >= t).mean() for t in grid])
    tf = np.array([(score[f & (y == 1)] >= t).mean() for t in grid])
    tot = pm[:, None] + pf[None, :]
    if mode == "eo":
        ok = np.abs(tf[None, :] - tm[:, None]) <= eo_tol
    else:
        ok = (arf[None, :] / np.maximum(arm[:, None], 1e-9)) >= dir_min
    tot = np.where(ok, tot, -np.inf)
    if not np.isfinite(tot).any():          # guard: never fall back silently to threshold (0, 0)
        raise ValueError(f"No threshold pair satisfies the '{mode}' fairness constraint on the
validation set")
    i, j = np.unravel_index(np.argmax(tot), tot.shape)
```

```
    return float(grid[i]), float(grid[j])


def apply_thresholds(score, g, t_m, t_f):
    return np.where(g == 1, score >= t_f, score >= t_m).astype(int)


# ---------------------------------------------------------------- one full experiment
def run_experiment(seed=42, make_extras=False):
    df = generate_applicants(seed=seed)
    strat = df["female"].astype(str) + df["repaid"].astype(str)
    tr, tmp = train_test_split(df, test_size=0.40, random_state=seed, stratify=strat)
    va, te = train_test_split(tmp, test_size=0.50, random_state=seed,
                              stratify=tmp["female"].astype(str) + tmp["repaid"].astype(str))
    sc = StandardScaler().fit(tr[FEATURES])
    X = {k: sc.transform(d[FEATURES]) for k, d in (("tr", tr), ("va", va), ("te", te))}
    y_tr, y_va, y_te = tr["repaid"].values, va["repaid"].values, te["repaid"].values
    g_tr, g_va, g_te = tr[PROTECTED].values, va[PROTECTED].values, te[PROTECTED].values

    res, decisions, extras = {}, {}, {}

    # M0: baseline - "fairness through unawareness" (gender excluded) + single global threshold
    base = LogisticRegression(max_iter=1000).fit(X["tr"], y_tr)
    s_va, s_te = base.predict_proba(X["va"])[:, 1], base.predict_proba(X["te"])[:, 1]
    t0 = best_global_threshold(s_va, y_va)
    decisions["Baseline LR"] = (s_te >= t0).astype(int)
    res["Baseline LR"] = audit(decisions["Baseline LR"], y_te, g_te, s_te); res["Baseline
LR"]["thresholds"] = [t0, t0]

    # M0b: gradient-boosted baseline (non-linear comparator)
    gb = HistGradientBoostingClassifier(max_iter=200, learning_rate=0.08,
random_state=seed).fit(X["tr"], y_tr)
    sg_va, sg_te = gb.predict_proba(X["va"])[:, 1], gb.predict_proba(X["te"])[:, 1]
    tg = best_global_threshold(sg_va, y_va)
    decisions["Baseline GBM"] = (sg_te >= tg).astype(int)
    res["Baseline GBM"] = audit(decisions["Baseline GBM"], y_te, g_te, sg_te); res["Baseline
GBM"]["thresholds"] = [tg, tg]

    # M1: pre-processing - reweighing
    w = reweighing_weights(g_tr, y_tr)
    rw = LogisticRegression(max_iter=1000).fit(X["tr"], y_tr, sample_weight=w)
    r_va, r_te = rw.predict_proba(X["va"])[:, 1], rw.predict_proba(X["te"])[:, 1]
    t1 = best_global_threshold(r_va, y_va)
    decisions["Reweighing"] = (r_te >= t1).astype(int)
    res["Reweighing"] = audit(decisions["Reweighing"], y_te, g_te, r_te);
res["Reweighing"]["thresholds"] = [t1, t1]

    # M2: post-processing - equal-opportunity group thresholds on baseline scores
    a, b = group_thresholds(s_va, y_va, g_va, "eo")
    decisions["EO thresholds"] = apply_thresholds(s_te, g_te, a, b)
    res["EO thresholds"] = audit(decisions["EO thresholds"], y_te, g_te, s_te); res["EO
thresholds"]["thresholds"] = [a, b]

    # M2b: post-processing - minimum-intervention DIR >= 0.80 thresholds
    a, b = group_thresholds(s_va, y_va, g_va, "dir")
    decisions["DIR>=0.80 thresholds"] = apply_thresholds(s_te, g_te, a, b)
    res["DIR>=0.80 thresholds"] = audit(decisions["DIR>=0.80 thresholds"], y_te, g_te, s_te)
    res["DIR>=0.80 thresholds"]["thresholds"] = [a, b]

    # M3: combined - reweighing + EO thresholds
    a, b = group_thresholds(r_va, y_va, g_va, "eo")
    decisions["Reweigh + EO"] = apply_thresholds(r_te, g_te, a, b)
    res["Reweigh + EO"] = audit(decisions["Reweigh + EO"], y_te, g_te, r_te); res["Reweigh +
EO"]["thresholds"] = [a, b]

    if make_extras:
        # proxy leakage: can gender be recovered from the 'gender-blind' features?
        Xa, Xb, ya, yb = train_test_split(X["tr"], g_tr, test_size=0.3, random_state=seed,
stratify=g_tr)
        leak = LogisticRegression(max_iter=1000).fit(Xa, ya)
        extras["proxy_auc"] = float(roc_auc_score(yb, leak.predict_proba(Xb)[:, 1]))
        extras["feature_gender_corr"] = {f: float(np.corrcoef(tr[f], tr["female"])[0, 1]) for f in
FEATURES}
```

```
        extras["feature_rural_corr"] = {f: float(np.corrcoef(tr[f], tr["rural"])[0, 1]) for f in
FEATURES}
        extras["lr_coef"] = dict(zip(FEATURES, map(float, base.coef_[0])))
        pi = permutation_importance(base, X["te"], y_te, n_repeats=10, random_state=seed,
scoring="roc_auc")
        extras["perm_importance"] = dict(zip(FEATURES, map(float, pi.importances_mean)))
        extras["calibration"] = {"male": {"mean_pred": float(s_te[g_te == 0].mean()), "actual":
float(y_te[g_te == 0].mean())},
                                 "female": {"mean_pred": float(s_te[g_te == 1].mean()), "actual":
float(y_te[g_te == 1].mean())}}
        extras["reweigh_weight_range"] = [float(w.min()), float(w.max())]
        extras["n"] = {"train": len(tr), "val": len(va), "test": len(te)}
        extras["base_rate_repay"] = {"male": float(y_te[g_te == 0].mean()), "female": float(y_te[g_te
== 1].mean())}
        extras["female_share"] = float(g_te.mean())
        extras["dir_ci_baseline"] = dir_bootstrap(decisions["Baseline LR"], g_te)
        extras["dir_ci_eo"] = dir_bootstrap(decisions["EO thresholds"], g_te)
        extras["dir_ci_reweigh_eo"] = dir_bootstrap(decisions["Reweigh + EO"], g_te)
        extras["dir_ci_dir80"] = dir_bootstrap(decisions["DIR>=0.80 thresholds"], g_te)
        # intersectional approval rates (gender x region), baseline vs EO thresholds
        inter = {}
        for name in ("Baseline LR", "EO thresholds", "Reweigh + EO"):
            d = decisions[name]
            inter[name] = {f"{'F' if gg else 'M'}-{'Rural' if rr else 'Urban'}":
                           float(d[(g_te == gg) & (te['rural'].values == rr)].mean()) for gg in (0, 1)
for rr in (0, 1)}
        extras["intersectional"] = inter
        # trade-off frontier: sweep the female threshold, hold male threshold at baseline optimum
        sweep = []
        for tf_ in np.linspace(0.40, 0.99, 60):
            d = apply_thresholds(s_te, g_te, t0, tf_); au = audit(d, y_te, g_te)
            sweep.append({"t_f": float(tf_), "DIR": au["DIR"], "EOD": au["EOD"], "profit":
au["profit"]})
        extras["frontier"] = sweep
        # robustness stress test: Gaussian noise injected into standardised test features (thresholds
frozen)
        rn = np.random.default_rng(seed + 99); stress = []
        tb, te_ = res["Baseline LR"]["thresholds"][0], res["EO thresholds"]["thresholds"]
        for sig in (0.0, 0.25, 0.5, 1.0):
            sn = base.predict_proba(X["te"] + rn.normal(0, sig, X["te"].shape))[:, 1]
            ab = audit((sn >= tb).astype(int), y_te, g_te, sn)
            ae = audit(apply_thresholds(sn, g_te, te_[0], te_[1]), y_te, g_te, sn)
            stress.append({"sigma": sig, "AUC": ab["AUC"], "DIR_base": ab["DIR"], "DIR_eo": ae["DIR"],
                           "EOD_base": ab["EOD"], "EOD_eo": ae["EOD"], "profit_base":
ab["profit_per_applicant"],
                           "profit_eo": ae["profit_per_applicant"]})
        extras["stress"] = stress
        extras["scores_te"] = (s_te, y_te, g_te)
        extras["data_summary"] = df.groupby("female")[FEATURES + ["repaid"]].mean().round(3).to_dict()
    return res, extras


# ---------------------------------------------------------------- figures
def make_figures(res, ex):
    order = ["Baseline LR", "Baseline GBM", "Reweighing", "EO thresholds", "DIR>=0.80 thresholds",
"Reweigh + EO"]
    labels = ["Baseline\nLR", "Baseline\nGBM", "Reweighing", "EO\nthresholds", "DIR≥0.80\nthresholds",
"Reweigh\n+ EO"]
    col = ["#7f7f7f", "#a6a6a6", "#4c78a8", "#2e8b57", "#8c6bb1", "#e07b39"]
    # Fig 5.1 approval rate by group
    fig, ax = plt.subplots(figsize=(6.4, 3.3)); xx = np.arange(len(order)); wd = 0.38
    ax.bar(xx - wd / 2, [res[k]["approval_male"] * 100 for k in order], wd, label="Male",
color="#4c78a8")
    ax.bar(xx + wd / 2, [res[k]["approval_female"] * 100 for k in order], wd, label="Female",
color="#e07b39")
    ax.set_xticks(xx); ax.set_xticklabels(labels, fontsize=8); ax.set_ylabel("Approval rate (%)");
ax.legend(frameon=False)
    plt.tight_layout(); plt.savefig("figs/fig_approval.png", dpi=200); plt.close()
    # Fig 5.2 DIR and TPR gap
    fig, axs = plt.subplots(1, 2, figsize=(6.6, 3.2))
    axs[0].bar(xx, [res[k]["DIR"] for k in order], color=col); axs[0].axhline(0.8, color="red",
ls="--", lw=1)
    axs[0].text(len(order) - 0.5, 0.82, "0.80 threshold", color="red", ha="right", fontsize=8)
```

```
    axs[0].set_xticks(xx); axs[0].set_xticklabels(labels, fontsize=6.5); axs[0].set_ylabel("Disparate
impact ratio"); axs[0].set_ylim(0, 1.15)
    axs[1].bar(xx, [abs(res[k]["EOD"]) for k in order], color=col)
    axs[1].set_xticks(xx); axs[1].set_xticklabels(labels, fontsize=6.5); axs[1].set_ylabel("|TPR gap|
(equal opportunity)")
    plt.tight_layout(); plt.savefig("figs/fig_dir_eod.png", dpi=200); plt.close()
    # Fig 5.3 profit vs DIR
    fig, ax = plt.subplots(figsize=(5.6, 3.5))
    fr = ex["frontier"]
    ax.plot([p["DIR"] for p in fr], [p["profit"] / 1e5 for p in fr], color="#999999", lw=1.2,
label="Female-threshold sweep")
    for k, c, lb in zip(order, col, ["Baseline LR", "Baseline GBM", "Reweighing", "EO thresholds",
"DIR≥0.80", "Reweigh+EO"]):
        ax.scatter(res[k]["DIR"], res[k]["profit"] / 1e5, s=55, color=c, zorder=3, label=lb)
    ax.axvline(0.8, color="red", ls="--", lw=1); ax.set_xlabel("Disparate impact ratio (female /
male)")
    ax.set_ylabel("Test-set portfolio profit (Rs. lakh)"); ax.legend(frameon=False, fontsize=7,
loc="lower right")
    plt.tight_layout(); plt.savefig("figs/fig_tradeoff.png", dpi=200); plt.close()
    # Fig 4.x proxy correlations
    fig, ax = plt.subplots(figsize=(5.8, 3.3)); fn = list(ex["feature_gender_corr"].keys())
    cg = [ex["feature_gender_corr"][f] for f in fn]; cr = [ex["feature_rural_corr"][f] for f in fn]; yy
= np.arange(len(fn))
    ax.barh(yy - 0.2, cg, 0.4, color="#e07b39", label="corr. with female"); ax.barh(yy + 0.2, cr, 0.4,
color="#4c78a8", label="corr. with rural")
    ax.set_yticks(yy); ax.set_yticklabels(fn, fontsize=8); ax.axvline(0, color="k", lw=0.6);
ax.set_xlabel("Pearson correlation"); ax.legend(frameon=False, fontsize=8)
    plt.tight_layout(); plt.savefig("figs/fig_proxy.png", dpi=200); plt.close()
    # Fig score distribution
    s, y, g = ex["scores_te"]; fig, ax = plt.subplots(figsize=(5.6, 3.1))
    ax.hist(s[g == 0], bins=40, alpha=0.6, density=True, color="#4c78a8", label="Male"); ax.hist(s[g ==
1], bins=40, alpha=0.6, density=True, color="#e07b39", label="Female")
    ax.axvline(res["Baseline LR"]["thresholds"][0], color="k", ls="--", lw=1); ax.set_xlabel("Predicted
repayment probability (baseline LR)"); ax.set_ylabel("Density"); ax.legend(frameon=False)
    plt.tight_layout(); plt.savefig("figs/fig_scores.png", dpi=200); plt.close()
    # Fig 3.1 pipeline
    fig, ax = plt.subplots(figsize=(6.6, 4.6)); ax.axis("off"); ax.set_xlim(0, 10); ax.set_ylim(0, 10)
    def box(x, y, w, h, t, c):
        ax.add_patch(plt.Rectangle((x, y), w, h, fc=c, ec="#333333", lw=1)); ax.text(x + w / 2, y + h /
2, t, ha="center", va="center", fontsize=8.8)
    def arr(x1, y1, x2, y2):
        ax.annotate("", xy=(x2, y2), xytext=(x1, y1), arrowprops=dict(arrowstyle="->", lw=1.1))
    box(0.2, 7.6, 2.6, 1.6, "Data layer\nSynthetic applicant\ngenerator (30,000)", "#dbe9f6"); box(3.7,
7.6, 2.6, 1.6, "Preprocessing\nlog-transform,\nscaling, 60/20/20 split", "#dbe9f6")
    box(7.2, 7.6, 2.6, 1.6, "Baseline models\nLR + GBM\n(gender excluded)", "#e8e8e8"); arr(2.8, 8.4,
3.7, 8.4); arr(6.3, 8.4, 7.2, 8.4)
    box(7.2, 4.6, 2.6, 1.8, "Bias audit module\nDIR, SPD, EOD, AOD,\nFPR gap, bootstrap CI",
"#fde5d3"); box(3.8, 4.6, 2.8, 1.8, "Proxy analysis\ncorrelation, gender-\nleakage AUC, perm. imp.",
"#fde5d3")
    box(0.2, 4.6, 3.0, 1.8, "Mitigation layer\nReweighing (pre)\nGroup thresholds (post)", "#d6ecd9");
arr(8.5, 7.6, 8.5, 6.4); arr(7.2, 5.5, 6.6, 5.5); arr(3.8, 5.5, 3.2, 5.5)
    box(0.2, 1.6, 3.0, 1.8, "Business evaluation\nprofit / loan, default\nrate in approved book",
"#f6e8b1"); box(3.8, 1.6, 2.8, 1.8, "Governance layer\nDPDP / RBI / EU AI Act\ncompliance + model
card", "#f6e8b1")
    box(7.2, 1.6, 2.6, 1.8, "Output\naudit_log.txt,\nresults.json, figures", "#dbe9f6"); arr(1.7, 4.6,
1.7, 3.4); arr(5.2, 4.6, 5.2, 3.4); arr(3.2, 2.5, 3.8, 2.5); arr(6.6, 2.5, 7.2, 2.5)
    ax.text(5, 0.5, "Human-in-the-loop review: loan officer overrides and adverse-action reasons are
logged", ha="center", fontsize=7.5, style="italic")
    plt.tight_layout(); plt.savefig("figs/fig_pipeline.png", dpi=200); plt.close()


# ---------------------------------------------------------------- main
if __name__ == "__main__":
    log = []
    def P(*a):
        s = " ".join(str(x) for x in a); print(s); log.append(s)
    P("=" * 78); P("FAIR CREDIT AUDIT - digital micro-lending (synthetic data, seed 42)"); P("=" * 78)
    res, ex = run_experiment(42, make_extras=True)
    P(f"Applicants: {N}  | split train/val/test =
{ex['n']['train']}/{ex['n']['val']}/{ex['n']['test']}")
    P(f"Female share (test): {ex['female_share']:.3f} | true repay rate  M:
{ex['base_rate_repay']['male']:.3f}  F: {ex['base_rate_repay']['female']:.3f}")
    P(f"Proxy leakage: gender recoverable from gender-blind features, AUC = {ex['proxy_auc']:.3f}")
    P("-" * 78)
```

```
P(f"{'Method':<22}{'AR_M':>7}{'AR_F':>7}{'DIR':>7}{'SPD':>8}{'TPR_M':>7}{'TPR_F':>7}{'EOD':>8}{'AUC':>7
}{'Profit/app':>12}")
    for k, v in res.items():

P(f"{k:<22}{v['approval_male']:>7.3f}{v['approval_female']:>7.3f}{v['DIR']:>7.3f}{v['SPD']:>8.3f}"
          f"{v['TPR_male']:>7.3f}{v['TPR_female']:>7.3f}{v['EOD']:>8.3f}{v.get('AUC',
float('nan')):>7.3f}{v['profit_per_applicant']:>12.1f}")
    P("-" * 78)
    for k in ("Baseline LR", "DIR>=0.80 thresholds", "EO thresholds", "Reweigh + EO"):
        P(f"Thresholds {k:<22} male={res[k]['thresholds'][0]:.2f}
female={res[k]['thresholds'][1]:.2f}")
    P("Bootstrap 95% CI of DIR (500 resamples):")
    for k, n_ in (("dir_ci_baseline", "Baseline LR"), ("dir_ci_dir80", "DIR>=0.80 thresholds"),
("dir_ci_eo", "EO thresholds"), ("dir_ci_reweigh_eo", "Reweigh + EO")):
        P(f"  {n_:<22} [{ex[k][0]:.3f}, {ex[k][1]:.3f}]")
    P(f"Calibration (mean predicted vs actual repay rate): male
{ex['calibration']['male']['mean_pred']:.3f}/{ex['calibration']['male']['actual']:.3f} | "
      f"female
{ex['calibration']['female']['mean_pred']:.3f}/{ex['calibration']['female']['actual']:.3f}")
    P(f"Reweighing weight range: {ex['reweigh_weight_range'][0]:.3f} -
{ex['reweigh_weight_range'][1]:.3f}")
    P("Noise stress test (sigma in standardised units; thresholds frozen):")
    for r_ in ex["stress"]:
        P(f"  sigma={r_['sigma']:.2f} AUC={r_['AUC']:.3f} | DIR base {r_['DIR_base']:.3f} -> EO
{r_['DIR_eo']:.3f} | "
          f"EOD base {r_['EOD_base']:+.3f} -> EO {r_['EOD_eo']:+.3f} | profit/app base
{r_['profit_base']:.0f} -> EO {r_['profit_eo']:.0f}")
    P("Intersectional approval rates:")
    for k, v in ex["intersectional"].items():
        P(f"  {k:<16}", {a: round(b, 3) for a, b in v.items()})
    P("Permutation importance (baseline LR, drop in AUC):")
    for f, v in sorted(ex["perm_importance"].items(), key=lambda t: -t[1]):
        P(f"  {f:<18}{v:.4f}   coef={ex['lr_coef'][f]:+.3f}
corr(female)={ex['feature_gender_corr'][f]:+.3f}")
    # robustness over 10 data-generation seeds
    P("-" * 78); P("Robustness: 10 independent synthetic datasets (seeds 1-10), mean +/- std")
    keys = ["Baseline LR", "Reweighing", "EO thresholds", "DIR>=0.80 thresholds", "Reweigh + EO"]
    agg = {k: {"DIR": [], "EOD": [], "AUC": [], "profit_per_applicant": []} for k in keys}
    for sd in range(1, 11):
        rr, _ = run_experiment(sd)
        for k in keys:
            for m_ in agg[k]:
                agg[k][m_].append(rr[k][m_])
    robust = {}
    for k in keys:
        robust[k] = {m_: [float(np.mean(v)), float(np.std(v))] for m_, v in agg[k].items()}
        robust[k]["DIR_min"] = float(np.min(agg[k]["DIR"])); robust[k]["seeds_DIR_ge_0.80"] =
int(np.sum(np.array(agg[k]["DIR"]) >= 0.80))
        P(f"{k:<22} (DIR>=0.80 in {robust[k]['seeds_DIR_ge_0.80']}/10 seeds, min
{robust[k]['DIR_min']:.3f}) DIR {robust[k]['DIR'][0]:.3f}+/-{robust[k]['DIR'][1]:.3f} | EOD
{robust[k]['EOD'][0]:+.3f}+/-{robust[k]['EOD'][1]:.3f} | "
          f"AUC {robust[k]['AUC'][0]:.3f}+/-{robust[k]['AUC'][1]:.3f} | profit/app
{robust[k]['profit_per_applicant'][0]:.0f}+/-{robust[k]['profit_per_applicant'][1]:.0f}")
    make_figures(res, ex)
    ex.pop("scores_te")
    json.dump({"results": res, "extras": ex, "robust": robust}, open("results.json", "w"), indent=1)
    open("audit_log.txt", "w").write("\n".join(log))
    P("Saved results.json, audit_log.txt, figs/*.png")
```

APPENDIX B

AUDIT LOGS, MODEL CARD & VERIFICATION

B.1 Complete audit log

Verbatim contents of audit_log.txt produced by the run reported in this document.

```
================================================================================
FAIR CREDIT AUDIT - digital micro-lending (synthetic data, seed 42)
================================================================================
Applicants: 30000  | split train/val/test = 18000/6000/6000
Female share (test): 0.423 | true repay rate  M: 0.813  F: 0.818
Proxy leakage: gender recoverable from gender-blind features, AUC = 0.897
--------------------------------------------------------------------------------
Method                   AR_M   AR_F    DIR     SPD  TPR_M  TPR_F     EOD    AUC  Profit/app
Baseline LR             0.556  0.400  0.718  -0.157  0.620  0.456  -0.164  0.745       246.3
Baseline GBM            0.571  0.433  0.758  -0.138  0.633  0.487  -0.146  0.734       229.6
Reweighing              0.495  0.329  0.665  -0.166  0.560  0.379  -0.181  0.745       253.3
EO thresholds           0.529  0.481  0.909  -0.048  0.595  0.544  -0.051  0.745       267.6
DIR>=0.80 thresholds    0.491  0.481  0.980  -0.010  0.555  0.544  -0.011  0.745       267.1
Reweigh + EO            0.495  0.479  0.967  -0.016  0.560  0.541  -0.020  0.745       269.1
--------------------------------------------------------------------------------
Thresholds Baseline LR            male=0.86 female=0.86
Thresholds DIR>=0.80 thresholds   male=0.88 female=0.83
Thresholds EO thresholds          male=0.87 female=0.83
Thresholds Reweigh + EO           male=0.88 female=0.83
Bootstrap 95% CI of DIR (500 resamples):
  Baseline LR            [0.682, 0.758]
  DIR>=0.80 thresholds   [0.932, 1.034]
  EO thresholds          [0.870, 0.959]
  Reweigh + EO           [0.922, 1.021]
Calibration (mean predicted vs actual repay rate): male 0.834/0.813 | female 0.786/0.818
Reweighing weight range: 0.987 - 1.018
Noise stress test (sigma in standardised units; thresholds frozen):
  sigma=0.00 AUC=0.745 | DIR base 0.718 -> EO 0.909 | EOD base -0.164 -> EO -0.051 | profit/app base 246 -> EO
268
  sigma=0.25 AUC=0.738 | DIR base 0.730 -> EO 0.915 | EOD base -0.158 -> EO -0.045 | profit/app base 245 -> EO
253
  sigma=0.50 AUC=0.732 | DIR base 0.716 -> EO 0.918 | EOD base -0.166 -> EO -0.047 | profit/app base 251 -> EO
250
  sigma=1.00 AUC=0.701 | DIR base 0.758 -> EO 0.932 | EOD base -0.141 -> EO -0.035 | profit/app base 179 -> EO
187
Intersectional approval rates:
  Baseline LR      {'M-Urban': 0.587, 'M-Rural': 0.525, 'F-Urban': 0.428, 'F-Rural': 0.371}
  EO thresholds    {'M-Urban': 0.559, 'M-Rural': 0.498, 'F-Urban': 0.517, 'F-Rural': 0.446}
  Reweigh + EO     {'M-Urban': 0.527, 'M-Rural': 0.463, 'F-Urban': 0.517, 'F-Rural': 0.442}
Permutation importance (baseline LR, drop in AUC):
  bill_ontime       0.0578   coef=+0.500   corr(female)=-0.000
  log_income        0.0243   coef=+0.344   corr(female)=-0.286
  upi_txn           0.0195   coef=+0.301   corr(female)=-0.312
  recharge_reg      0.0161   coef=+0.299   corr(female)=+0.002
  log_device_value  0.0012   coef=-0.078   corr(female)=-0.363
  addr_stability    0.0004   coef=+0.036   corr(female)=-0.209
  contacts          0.0002   coef=+0.042   corr(female)=-0.560
  age               -0.0001   coef=-0.034   corr(female)=-0.003
  formal_job        -0.0001   coef=+0.007   corr(female)=-0.178
--------------------------------------------------------------------------------
Robustness: 10 independent synthetic datasets (seeds 1-10), mean +/- std
Baseline LR            (DIR>=0.80 in 0/10 seeds, min 0.670) DIR 0.738+/-0.030 | EOD -0.159+/-0.015 | AUC
0.751+/-0.006 | profit/app 276+/-14
Reweighing             (DIR>=0.80 in 0/10 seeds, min 0.666) DIR 0.740+/-0.031 | EOD -0.158+/-0.017 | AUC
0.751+/-0.006 | profit/app 275+/-13
EO thresholds          (DIR>=0.80 in 10/10 seeds, min 0.931) DIR 0.984+/-0.040 | EOD -0.010+/-0.021 | AUC
0.751+/-0.006 | profit/app 282+/-17
DIR>=0.80 thresholds   (DIR>=0.80 in 10/10 seeds, min 0.834) DIR 0.937+/-0.082 | EOD -0.036+/-0.047 | AUC
0.751+/-0.006 | profit/app 281+/-14
Reweigh + EO           (DIR>=0.80 in 10/10 seeds, min 0.923) DIR 0.990+/-0.035 | EOD -0.007+/-0.020 | AUC
0.751+/-0.006 | profit/app 282+/-18
```

B.2 Model card

Format follows the model-card framework of [23].

Table B.1: Model card for the micro-lending credit-scoring pipeline

Section	Content
Model details	Logistic regression (L2, standardised inputs) producing a repayment probability; version 1.0 (academic prototype); optional post-processing with group-specific thresholds
Intended use	Demonstration of bias auditing and mitigation for unsecured ₹10,000 digital micro-loans. **Not** for production lending decisions
Out-of-scope use	Real credit decisions; any population other than the synthetic one; pricing; collections
Training data	18,000 synthetic applicants (60% split); 9 features; gender and region excluded from inputs
Evaluation data	6,000 synthetic test applicants; 42.3% female; 10 additional independently generated datasets
Metrics	AUC; approval rate, DIR, SPD, EOD, AOD by gender; intersectional approval by gender × region; profit per applicant; noise-stress results
Quantitative analysis: baseline	AUC 0.745; DIR 0.718; EOD −0.164; profit ₹246 per applicant
Quantitative analysis: mitigated (EO thresholds)	AUC 0.745; DIR 0.909; EOD −0.051; profit ₹268 per applicant
Ethical considerations	Gender is inferable from model inputs (AUC 0.897); feature store must be handled as sensitive data. Group thresholds use gender at decision time, which may be unlawful in some jurisdictions
Caveats and recommendations	Synthetic data; binary gender; no reject inference; re-audit on real data before any use; monitor DIR monthly; keep human review near the threshold

B.3 Verification sheet

Table B.2: Verification checklist

Check	Result	Evidence
Protected attribute excluded from model inputs	Pass	assert in source; Section 4.3
Thresholds tuned on validation, metrics on test	Pass	run_experiment(); Section 3.1
Fixed seeds; full run reproducible	Pass	seeds 42 and 1–10; Appendix A
Baseline meets DIR ≥ 0.80	**Fail (the finding)**	DIR 0.718; 0/10 datasets
Threshold mitigations meet DIR ≥ 0.80 in all datasets	Pass	Table 5.7
Reweighing alone meets DIR ≥ 0.80	**Fail**	DIR 0.665; 0/10 datasets
|EOD| ≤ 0.02 on test for seed 42 (EO thresholds)	**Not met**	EOD −0.051 (validation target met; mean over 10 datasets −0.010)

Check	Result	Evidence
Intersectional ratio F-Rural / M-Urban ≥ 0.80 after EO	**Marginally below**	0.798; Table 5.8
Graphical dashboard demonstrated	Not done	Future work, Section 7.3